Full-Time Schooling and Student Outcomes: Identification under Compositional Changes

Andreas Azambuja Barbisan*

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Abstract

This study estimates proficiency gains associated with school adoption of the Full-Day Education Program (PEI), a policy that expanded school hours and reorganized management in São Paulo state schools. The research exploits the staggered rollout of the program using a difference-in-differences framework, comparing Two-Way Fixed Effects (TWFE), Callaway and Sant'Anna (2021), Sant'Anna and Zhao (2020), and an aggregate approximation inspired by Sant'Anna and Xu (2026). The main estimates indicate average post-treatment gains between approximately 0.50 and 0.68 standard deviations in standardized Portuguese Language and Mathematics exams. The inclusion of composition covariates constructed from enrollment microdata and municipal controls does not substantially alter the estimates, indicating that aggregate observable composition is not sufficient, in the estimated specifications, to eliminate the gains. The interpretation, however, is limited: the absence of an individual link between enrollment and test scores prevents a full decomposition of learning, selection, and test participation, so the results should be read as aggregate evidence conditional on the identifying assumptions adopted.

Keywords: Economics of Education, Difference-in-differences, Full-day schooling

JEL Codes: I21, I28, J24

* **Andreas Azambuja Barbisan**: Insper Institute of Education and Research, Rua Quatá, 300, São Paulo, SP, Brazil 04546-042. E-mail: andreasab1@insper.edu.br

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1 Introduction

According to a report released by the Organisation for Economic Co-operation and Development (OECD) in December 2023, Brazil remained below the average of OECD countries in PISA (*Programme for International Student Assessment*), even though the country did not experience a statistically significant decline once the impact of the pandemic on Brazilian education is taken into account (OECD, 2023). Learning deficits remain a long-standing bottleneck for the country's development and continue to motivate debates over the appropriate design of education policy.

In this context, policies that extend the school day have become increasingly prominent. In the state of Sao Paulo, the main example of this policy family is the Full-Time Education Program (*Programa Ensino Integral*, PEI), established in 2012 (Estado de São Paulo, 2012). The program combines longer school hours with new management practices and the construction of each student's life project. Through staggered adoption, PEI expanded substantially over the last decade. Estimating the gains associated with policies such as PEI is important, but it also raises complex methodological challenges because of the way the policy was implemented.

The starting point of this paper is the empirical problem of estimating the gains associated with PEI on student performance in the Sao Paulo State School Performance Assessment System (SARESP). In a recent paper, Scorzafave et al. (2023) evaluated the program using the difference-in-differences (DiD) frontier estimator of Callaway and Sant'Anna (2021) (CS21), overcoming the limitations of the traditional two-way fixed effects (TWFE) estimator, which, as shown by Goodman-Bacon (2021) and Chaisemartin and D'Haultfœuille (2020), can make "forbidden comparisons" and generate severely biased estimates under staggered adoption and heterogeneous treatment effects.

The structure of educational assessment data makes this problem even harder. SARESP exams do not follow the same student over time, forming repeated cross-sections (RCS). In addition, the empirical literature documents that converting a school to the PEI model is associated with changes in the composition of its student body (Favaro, 2023) and with spillover effects on nearby regular schools (Santos, 2022). In short, the policy may change not only the treatment status of units, but also who appears in the sample after treatment. When this explicit compositional change occurs, most modern DiD estimators for RCS data, including Callaway and Sant'Anna (2021), rely on a stationarity assumption that may be strained, attributing to the program proficiency differences that could also reflect student selection (Sant'Anna and Xu, 2026).

Against this background, this paper goes beyond replicating the empirical exercise in Scorzafave et al. (2023). The central objective is to use PEI as an empirical laboratory to discuss which DiD estimator is most appropriate for this type of data, presenting a progression of estimators that address the structural problems with increasing rigor. I re-estimate the gains associated with PEI comparatively, moving through four approaches. First, I use the classical TWFE specification as a benchmark, while recognizing that it assumes homogeneous effects and fails structurally under staggered timing. Second, I apply the estimator of Callaway and Sant'Anna (2021), which solves the timing and temporal heterogeneity problem by estimating $ATT(g, t)$, replicating the baseline strategy of Scorzafave et al. (2023). Third, because this method still imposes stationarity and does not correct for compositional changes in the cross-section, I use the doubly robust DiD estimator of Sant'Anna and Zhao (2020) (DR-DiD), which combines outcome regression and weighting to handle covariates. Finally, I implement an aggregate approximation inspired by Sant'Anna and Xu (2026) (S&X). This approximation does not fully reproduce the individual-level estimator proposed by the authors, because the available data do not allow individual enrollment records to be linked to individual SARESP scores; nevertheless, it allows me to assess whether incorporating observable compositional changes at the school-grade level changes the magnitude of the estimated effects.

The empirical strategy exploits the staggered adoption of the program across the state of Sao Paulo. I approximate the intuition of the Sant'Anna and Xu (2026) estimator to evaluate whether observable compositional changes alter the magnitude of the estimated effects. Pre-treatment dynamics are consistent with the parallel trends assumption in the preferred specifications, although they do not formally prove it. The estimates indicate that program adoption is associated with proficiency gains in standardized exams, both in Portuguese Language and Mathematics, for the two grades analyzed: 9th grade of elementary school and the 3rd grade of high school.

These results, however, do not fully separate learning gains, student selection, exam participation, and possible spillovers onto nearby regular schools. Because the data required to link enrollment records to exam microdata are unavailable, unlike the approach in Scorzafave et al. (2023), the microdata are used only to construct aggregate covariates for the estimators. This limits the inferences that can be drawn from the results.

The contribution of this paper is to show why compositional changes matter for evaluations using RCS data, a common format in the economics of education and in standardized test databases, and to discuss through a transparent empirical adaptation whether the magnitude of the results changes when the observable composition of the student body is incorporated into estimation. The paper therefore contributes in two ways. First, in the economics of education, it re-evaluates PEI using estimators suited to staggered adoption and aggregate controls for student composition. Second, in applied causal inference, it makes explicit the distance between the ideal estimator, which would require perfectly linked microdata, and the feasible estimator in the available administrative setting.

The remainder of the paper is structured as follows. Section 2 reviews the literature on full-time schooling, the structure of PEI, and the evolution of DiD estimators. Section 3 presents the microeconomic model underlying the program's channels of action. Section 4 describes the data and the theoretical intuition and assumptions behind the progression of econometric methods. Section 5 presents the estimation results, contrasts the estimators, and discusses the test for compositional changes. Finally, Section 6 concludes.

2 Literature Review

This section is divided into three parts. Section 2.1 reviews the literature on full-time schooling programs, particularly in the Brazilian and South American contexts. Section 2.2 describes the PEI in the state of Sao Paulo, detailing its staggered implementation and discussing recent findings on student composition. Finally, Section 2.3 discusses the evolution of difference-in-differences estimators and motivates the methodological choices in this paper.

2.1 Full-Time Education

Increasing instructional time is often cited as a central lever for improving educational quality. One of the first papers to address this subject in the South American context is Bellei (2009), which evaluated a full-time school program in Chile. Using a Difference-in-Differences methodology, the author found positive and heterogeneous effects, with the program generating a greater impact in rural areas, among public school students, and students already at the top of the performance distribution.

In the Brazilian context, however, the literature demonstrates that results vary drastically depending on the design and depth of the educational policy adopted. Almeida et al. (2016) evaluated the federal program *"Mais Educação"*, which focused on providing complementary extracurricular activities, increasing the daily school load to a minimum of 7 hours. Despite the extra funding, the authors found a null impact on standardized test scores in Portuguese and on school dropout rates, and even a negative impact on Mathematics. This result suggests that simple extension of school hours, without profound pedagogical restructuring, is insufficient to leverage learning.

On the other hand, more structured state programs have shown success. In Pernambuco, Rosa et al. (2022) found significant gains in Mathematics and Language (approximately 0.22 and 0.19 standard deviations) resulting from the state full-time education program. Complementarily, Araujo et al. (2020) showed that this same program positively affected scores on the National High School Exam (ENEM), highlighting a "multiplier effect" of hours spent on complementary activities.

In the State of São Paulo, the precursor experience of the *"Programa Escola de Tempo*

Integral" (ETI), implemented between 2007 and 2008, illustrates the difficulties of school day extension policies. Aquino (2011) analyzed the ETI using panel data and Propensity Score Matching, finding essentially null results in Mathematics and passing rates. The author points out that the lack of activity planning focused on proficiency and coordination problems motivated the initiative's failure. It is evident, therefore, that there was a need for a program that combined classroom time with a new management and teaching architecture gap that would later be filled by the PEI.

2.2 The Integral Education Program (PEI) in SÃ£o Paulo

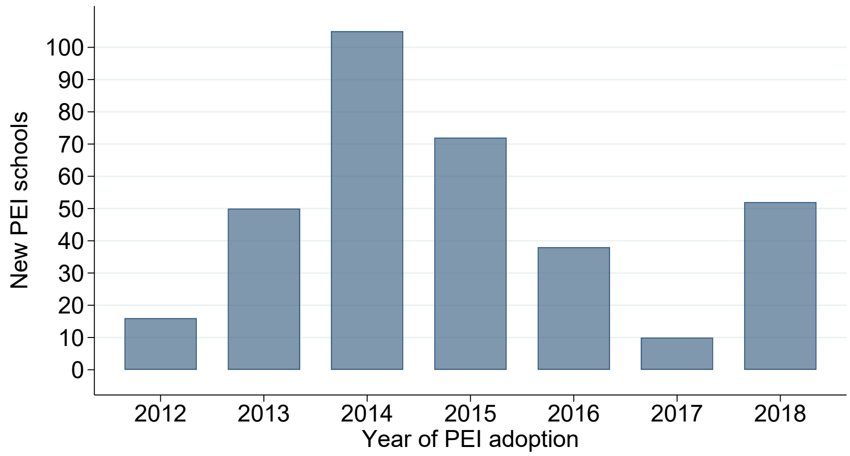
Established in 2012, the Integral Education Program (PEI) emerged as the main bet of the São Paulo state network to overcome the bottlenecks of previous models. The formulation of the PEI distinguishes itself by proposing a full-time school that combines extended stay (8 to 9 hours daily) with a profound redesign of the pedagogical project and the management model (Estado de São Paulo, 2012).

To operationalize this proposal, converted schools adopted innovations such as the *Life Project* (where students structure their long-term goals), Electives, Experimental Practices, and Tutoring. From a management perspective, the program introduced the PDCA (Plan, Do, Check, and Act) methodology and established the Full and Integral Dedication Regime (RDPI), which guarantees a 75% salary bonus for teachers and managers in exchange for exclusive dedication to a single school unit, preventing the faculty from splitting their time between multiple schools (Estado de São Paulo, 2012). The implementation of the PEI occurred in a non-random and staggered manner over time. As illustrated in Figure 1, the program began with a restricted group of schools in 2012, increasing substantially over the years.

Figure 2 highlights the magnitude of this rollout, showing the consolidation of the PEI as a large-scale policy in the São Paulo educational system. Conditional adherence based on infrastructure criteria in the early years reinforces the character of the PEI as a change in a complex institutional arrangement, rather than a mere exogenous time shock.

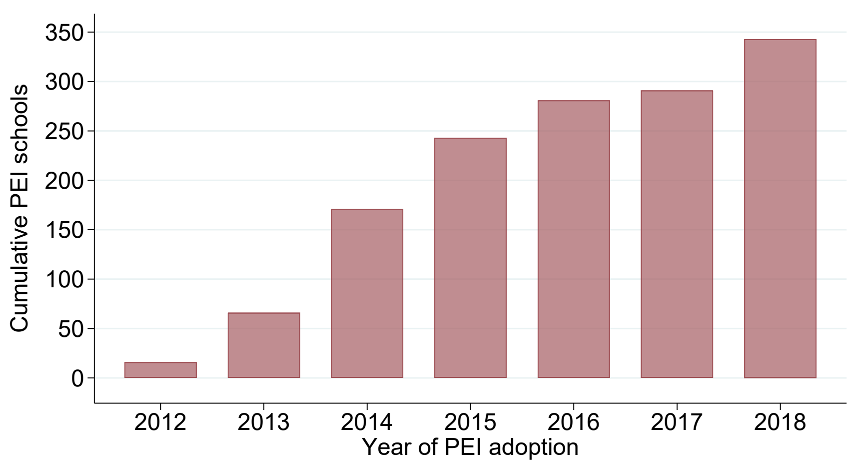
Early empirical evaluations of the PEI attest to its success in raising human capital. Fukushima, Quintão, and Pazello (2022) applied Difference-in-Differences to estimate the im-

Figure 1. New PEI schools by year of adoption



Source: SP Education Open Data. Author's elaboration.

Figure 2. Cumulative number of PEI schools in the state of Sao Paulo



Source: SP Education Open Data. Author's elaboration.

pact on SAEB results, reporting gains of about 0.46 standard deviations in Mathematics and Portuguese. Additionally, Kawahara (2019) found positive impacts on ENEM performance, *Prova Brasil* (a biennial assessment applied to students in public schools), and school flow indicators. More recently, Scorzafave et al. (2023) directly evaluated the impact on SARESP High School scores, finding gains of 0.31 and 0.21 standard deviations in Mathematics and Portuguese, respectively.

Despite the strong consensus surrounding positive direct results, a recent and critical literature has raised structural concerns regarding how the program affects the state network. Favaro (2023) analyzed compositional changes associated with program adoption and documented that the conversion to PEI actively alters the profile of the student body: the program increased the fraction of girls and reduced the percentage of non-white students and students with school age-grade distortion. Furthermore, a reduction in the total number of enrollments in newly converted schools was identified.

Simultaneously, Santos (2022) documented the existence of spillover effects on regular schools in the vicinity of PEI schools, showing that student migration alters the composition of neighboring schools, harming their performance. These two empirical findingsâ€”compositional change in treated units and contagion in the control groupâ€”form the basis of the problem that this work aims to solve, requiring a transition to the frontier of causal inference econometrics.

2.3 Methodological Evolution: The Evolution of Difference-in-Differences Estimators

Impact evaluation of public policies has gained centrality in contemporary economic research. As documented by Currie, Kleven, and Zwiers (2020), the discipline underwent an empirical turn in recent decades: applied microeconomics became a growing share of top journals, and quasi-experimental methods, especially difference-in-differences, came to organize a large share of causal evidence in public policy.

Despite this popularity, evaluations of public policies with multiple periods, such as the PEI rollout, historically relied almost exclusively on TWFE estimators. A recent econometric literature shows that this estimator can fail severely when the policy is adopted in a staggered manner and treatment effects are heterogeneous over time.

Goodman-Bacon (2021) shows that TWFE can make “forbidden comparisons,” using units already treated in earlier periods as controls for units treated later. Complementarily, Chaisemartin and D’Haultfœuille (2020) show that this contamination may generate strictly negative aggregation weights, potentially producing estimates with the opposite sign of the true policy effect.

To address the limitations of the classical model, the literature developed new estimators. The approach of Callaway and Sant’Anna (2021) became a primary solution for staggered adoption, isolating clean comparisons that use only not-yet-treated or never-treated units. In addition, Sant’Anna and Zhao (2020) advanced doubly robust estimators that combine regression adjustment and weighting (IPW) to handle covariates efficiently.

Finally, the data structure of educational assessments imposes an additional identification challenge. Exams such as SARESP form repeated cross-sections. The concern with DiD under compositional change appears early in Hong (2013), who shows in another empirical context that the composition of observed groups may change over time and bias conventional estimators. Murto (2024) revisits this argument and shows that propensity-score reweighting strategies can be adapted to handle compositional shifts in repeated cross-sections in a simpler way. As Sant’Anna and Xu (2026) emphasize, DiD estimators for this type of data usually rely on a stationarity assumption: the distribution of observed individuals in treated and control units should not change differentially because of treatment.

As Favaro (2023) documents for PEI, program adoption changes the composition of the student body in a statistically significant way. When this premise is violated by the policy itself, estimators that do not correct for composition may confuse student selection with learning gains, in addition to migration across schools. This paper starts from that concern, but adapts it to the publicly available data: instead of fully applying the individual-level estimator of Sant’Anna and Xu (2026), I use composition covariates aggregated by school, year, and grade to evaluate the sensitivity of the estimates to observable changes in the student body.

3 Microeconomic Model

This paper is based on a microeconomic model of an educational production function, in which a student's academic performance is the result of a combination of different inputs throughout the learning process. The approach follows the classical literature on the economics of education, in which learning is treated as the product of individual, family, school, and social factors (Hanushek, 1979).

Formally, the academic outcome of student i in period t, denoted by A_{it}, can be described by the following educational production function:

$$A_{it} = f\left(B_i^{(t)}, P_i^{(t)}, S_i^{(t)}, I_i\right). \tag{1}$$

In this expression, $B_i^{(t)}$ represents the student's family *background*, which includes socioeconomic characteristics of the household, cultural capital, and educational support conditions. $P_i^{(t)}$ corresponds to the set of peers with whom the student interacts in the school environment, reflecting possible educational externalities arising from the composition of the class or school.

The term $S_i^{(t)}$ represents the school inputs available to the student. In this paper, these inputs are modeled explicitly as a function of three main dimensions of the educational environment:

$$S_i^{(t)} = S(\tau, \pi, \sigma)_i^{(t)}, \tag{2}$$

where τ represents the time the student spends in school, π corresponds to the personal and socio-emotional development provided by the school environment, and σ represents the infrastructure and pedagogical resources available. Thus, school inputs are understood as a combination of these three factors, which jointly affect the learning process.

Finally, I_i represents the student's innate abilities—that is, relatively persistent individual characteristics that influence their learning capacity.

It is assumed that all these factors positively impact the student's academic outcome, so that the partial derivatives of the production function are positive. Formally,

$$\frac{\partial A_{it}}{\partial B_i^{(t)}} > 0,$$

$$\frac{\partial A_{it}}{\partial P_i^{(t)}} > 0,$$

$$\frac{\partial A_{it}}{\partial S(\tau, \pi, \sigma)_i^{(t)}} > 0,$$

$$\frac{\partial A_{it}}{\partial I_i} > 0.$$

Family *background* $B_i^{(t)}$ positively impacts performance, as a better family context is associated with better educational support conditions, which raises the student's academic outcome. Peers $P_i^{(t)}$ also exert a positive effect, since learning presents positive externalities: better peers tend to generate an environment more conducive to learning, raising individual performance.

School inputs $S(\tau, \pi, \sigma)_i^{(t)}$ positively affect the result and are composed of three dimensions. First, the time spent in school τ has a positive effect, i.e.,

$$\frac{\partial A_{it}}{\partial \tau} > 0,$$

because the more time dedicated to schooling, the greater the accumulation of knowledge tends to be. Second, personal development π also presents a positive effect,

$$\frac{\partial A_{it}}{\partial \pi} > 0,$$

since greater emotional and social stability favors the learning process. Finally, school infrastructure σ contributes positively to performance,

$$\frac{\partial A_{it}}{\partial \sigma} > 0,$$

as better physical and pedagogical resources expand the learning channels available to the student.

Additionally, innate abilities I_i positively influence academic performance,

$$\frac{\partial A_{it}}{\partial I_i} > 0,$$

since higher levels of individual aptitude tend to raise a given student's learning capacity.

3.1 Channels of action of the PEI

In the context of the integral model, the main channels of impact on academic performance are peers and school factors. Regarding peers $P_i^{(t)}$, full-time schools may attract more engaged students with better prior performance, while simultaneously potentially pushing away more vulnerable students or those who need to balance study and work. From a theoretical point of view, this dynamic raises the expected average performance, i.e.,

$$\frac{\partial A_{it}}{\partial P_i^{(t)}} > 0.$$

However, from an empirical and public policy evaluation standpoint, this same dynamic represents exactly the compositional-change bias discussed in recent literature (Favaro, 2023). This phenomenon requires econometric tools that reduce the risk of confusing the accounting effect generated by student turnover with learning gains, which underpins the methodological choices detailed in the next section.

Furthermore, school factors $S_i^{(t)}$, which can be decomposed into time spent τ, personal development π, and infrastructure σ, also constitute relevant channels of direct impact of the program. Regarding the time spent in school, it is expected that

$$\frac{\partial A_{it}}{\partial \tau} > 0,$$

since greater exposure to the school environment tends to increase the accumulation of learning, although, in isolation, this channel may eventually present a limited effect if it is not accompanied by pedagogical changes, as seen in the experience of Almeida et al. (2016).

With respect to personal development π, a positive effect on performance is expected,

$$\frac{\partial A_{it}}{\partial \pi} > 0,$$

as greater emotional, social, and behavioral stability, stimulated by PEI components such as the "Life Project" and tutoring, favors the continuity and quality of the learning process Araujo et al. (2020).

Finally, the improvement in school infrastructure σ expands pedagogical possibilities and available teaching methods, generating new forms of learning. Thus, it is expected that

$$\frac{\partial A_{it}}{\partial \sigma} > 0.$$

In summary, the integral model acts through two distinct channels on academic performance. The direct channel operates via the strengthening of school inputs (time spent τ, personal development π, and infrastructure σ), representing the learning gain that the policy intends to generate. The indirect channel operates via the alteration of the composition of peers $P_i^{(t)}$: by imposing an extended school day and a more demanding pedagogical model, the PEI selectively alters who remains enrolled in the treated school, attracting more engaged students and pushing away the most vulnerable.

From an impact evaluation standpoint, this distinction is fundamental. The parameter of interest in this paper, the Average Treatment Effect on the Treated (ATT), approximates the direct channel that the policy seeks to produce: the learning gain associated with changes in school time and organization. The peer channel, however, represents a composition problem. The observed increase in average proficiency among PEI schools may partially reflect a change in the student profile, not only pedagogical improvement. Formally, when the composition of $P_i^{(t)}$ changes systematically between the pre- and post-treatment periods, the stationarity assumption is violated, and estimators that do not correct for this change may confuse the compositional effect with the effect attributed to the policy.

4 Data and Methodology

This section describes the data and estimation strategy. Section 4.1 describes the data in detail. Section 4.2 presents descriptive evidence. Section 4.3 details the estimation strategy used in this paper, contrasting it with the estimator I attempted to approximate and showing where the analysis remains limited. Finally, Section 4.4 discusses threats to identification.

4.1 Data

The main outcome data come from the Sao Paulo State School Performance Assessment System (SARESP), made available by the Sao Paulo State Department of Education.[1] SARESP is an annual assessment administered in almost all schools in the state system, measuring proficiency in Portuguese Language and Mathematics. This paper uses 9th grade of elementary school and the 3rd grade of high school, two stages that are comparable over the period analyzed and relevant for assessing the accumulated effects of the policy. SARESP has its own classification of proficiency levels; the details are available in the original monograph appendix.

The unit of analysis is the school-year-grade-subject panel. This choice follows from the available administrative structure: enrollment microdata allow me to characterize school composition, but they do not reliably link each enrolled student to her individual SARESP score. This occurs because the way student identifiers are coded in the outcomes database differs from the coding used in the enrollment database. Therefore, the main estimation is aggregated at the school level, although it is informed by statistics constructed from the microdata.

The main window runs from 2011 to 2018. The year 2011 provides the period immediately before PEI began in 2012. The upper limit of 2018 preserves a pre-pandemic window and avoids the comparability break associated with COVID-19 and the interruption of SARESP in 2020. Schools that joined PEI only after 2018 are treated as not-yet-treated within the main window.

[1] The full databases are available at the Education Open Data portal: https://dados.educacao.sp.gov.br/.

The outcome is average school proficiency in each year, grade, and subject. To facilitate comparisons across years, scores are standardized within each year:

$$Y_{it}^{std} = \frac{Y_{it} - \bar{Y}_t}{\mathrm{sd}(Y_t)}, \tag{3}$$

where Y_{it} is the average proficiency of school i in year t, $\bar{Y}_t$ is the annual mean, and $\mathrm{sd}(Y_t)$ is the annual standard deviation. Estimated coefficients can therefore be interpreted in standard deviations of the annual proficiency distribution.

In addition to the proficiency panel, the paper uses enrollment microdata to construct aggregate composition covariates by school, year, and grade. These covariates include the share of female students, the share of students with a registered special need, average age, the share of Black or mixed-race students, the share of students with declared race/color, the share of Bolsa Familia beneficiaries, and school size measured by enrollment. The special-needs variable is an indicator equal to one when the administrative record reports any disability or specific educational condition. The data dictionary includes the following categories: multiple disability, blindness, low vision, severe or profound deafness, deafblindness, physical disability associated with cerebral palsy, physical disability-wheelchair user, Down syndrome, intellectual disability, monocular vision, childhood autism, Asperger syndrome, Rett syndrome, childhood disintegrative disorder, and giftedness.

When necessary, missing values in composition covariates are imputed using annual means. This choice preserves panel coverage and avoids making the main estimation depend only on schools with more complete administrative records. Section 5 presents a sensitivity diagnostic without imputation to document the sample cost of this decision.

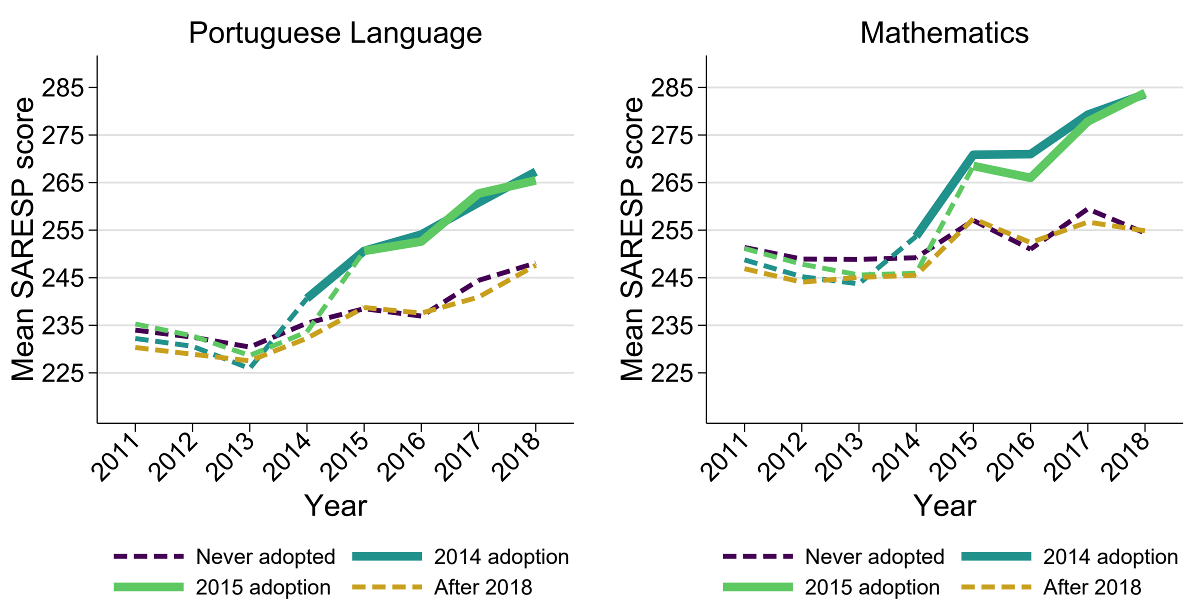
This construction reflects the main empirical limitation of the paper. The microdata are not used to estimate individual PEI effects, because the link between enrollment and individual scores is not reliable over the relevant period. The enrollment and SARESP outcome databases do not share the same student identifier, preventing a direct merge that would perfectly control for migration effects. Such a merge would only be possible with authorized administrative access from the Department of Education, as in Scorzafave et al. (2023). In this paper, the microdata enter as a source of aggregate covariates, allowing me to approximate observable compositional changes at the school level, but not at the student level.

I also incorporate municipal controls from local-characteristics databases made available by IBGE.[2] The municipal covariates are municipal GDP per capita, total municipal population, and the share of the population aged 0 to 19.

4.2 Descriptive Evidence

Before the causal estimation, Figures 3 and 4 show the evolution of average SARESP scores by school stage. They indicate that schools that join PEI experience proficiency growth after conversion. This evidence is useful for visualizing the basic pattern in the data, but it does not identify the causal effect of the program. The observed increase may reflect learning, pre-existing differences across schools, selection into the program, and changes in the composition of assessed students.

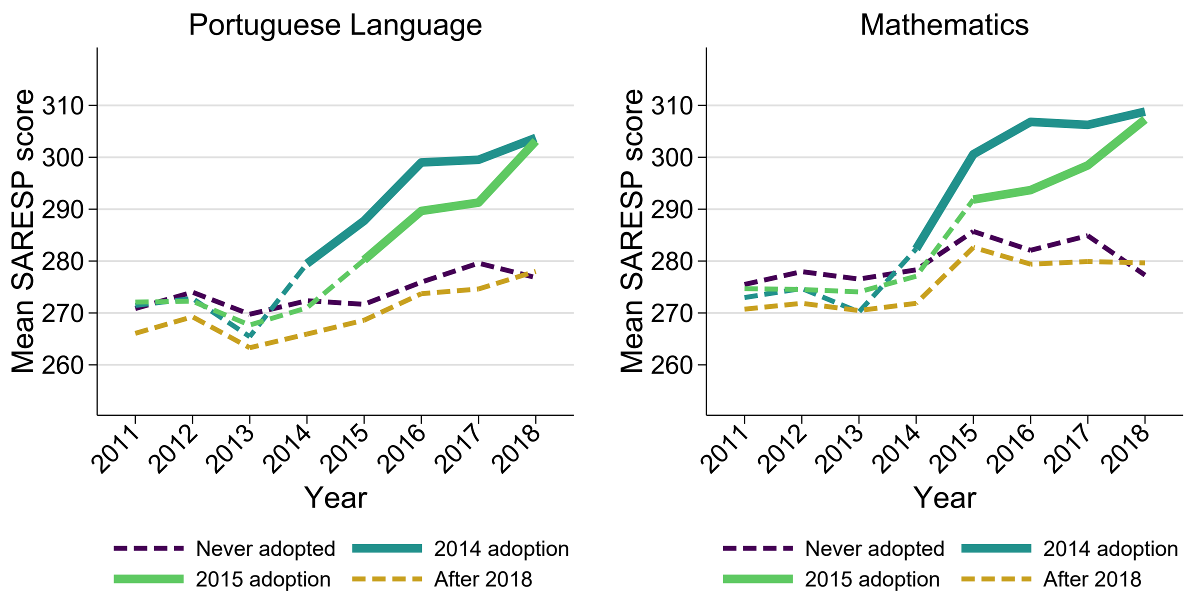
Figure 3. Evolution of SARESP scores - 9th grade



Source: Education Open Data, author's elaboration.

[2]Available at https://www.ibge.gov.br/estatisticas/economicas/contas-nacionais/9088-produto-interno-bruto-dos-municipios.html and https://www.ibge.gov.br/estatisticas/sociais/populacao/9103-estimativas-de-populacao.html.

Figure 4. Evolution of SARESP scores - 3rd grade of high school



Source: Education Open Data, author's elaboration.

The figures also help interpret the scale of the problem. In absolute terms, PEI schools appear to move toward higher levels of proficiency, especially in high school. However, visual comparisons of raw means do not separate the direct pedagogical channel from the compositional channel. A school may improve because it teaches better, but also because it begins to receive, retain, or assess a different set of students. This distinction is precisely what motivates the empirical strategy below.

For this reason, in the main estimation proficiency is standardized by year and the effects are interpreted in standard deviations. This choice prevents changes in the annual SARESP scale or distribution from contaminating comparisons across periods. The empirical object is not only whether the mean of PEI schools increased, but whether this increase remains when compared with not-yet-treated or never-treated schools in a structure compatible with staggered adoption.

4.3 Empirical Strategy

The empirical strategy exploits the staggered adoption of PEI across Sao Paulo state schools. Let G_i be the first year in which school i joins the program. The basic parameter of interest is the average treatment effect for cohort g in period t:

$$ATT(g,t) = E\left[Y_{it}(g) - Y_{it}(0) \mid G_i = g\right], \quad t \geq g, \tag{4}$$

where $Y_{it}(g)$ is the potential outcome of the school if treated from year g onward, and $Y_{it}(0)$ is the potential outcome without treatment. The central problem is that $Y_{it}(0)$ is not observed for schools that have already converted. Identification therefore depends on constructing a plausible counterfactual using schools that have not yet joined or that do not join the program within the analysis window.

Table 1 summarizes the progression of estimators used in the paper. The logic is not to present four equivalent answers. The sequence starts from a traditional benchmark, corrects the staggered-adoption problem, adds observable covariates, and finally tests the sensitivity of the results to compositional change.

From this point on, i indexes schools, not students as in the microeconomic model in Section 3. The first specification is an event study with school and year fixed effects. Let $k = t - G_i$ denote relative time to conversion; α_i represents school fixed effects, λ_t represents year fixed effects, and X_{it} contains aggregate covariates used as controls:

$$Y_{it}^{std} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbb{I}\{t - G_i = k\} + X_{it}'\gamma + \varepsilon_{it}. \tag{5}$$

This specification is useful as a reference, but it is not the preferred estimator. When adoption is staggered and treatment effects are heterogeneous over time, TWFE may compare already-treated schools with schools treated later, assigning weights to the average effect that are difficult to interpret.

The second specification uses the estimator of Callaway and Sant'Anna (2021). The method separately estimates $ATT(g,t)$ for each cohort and period, using not-yet-treated or never-treated schools as the comparison group. This prevents schools already exposed to PEI from serving as controls for schools treated later. The dynamic aggregation of these

Table 1. Role of the estimators in the empirical strategy

Estimator	Comparison used	What it corrects	Main limitation
TWFE	Event study with school and year fixed effects	Fixed differences across schools and common year shocks	May make forbidden comparisons under staggered adoption
CS21	$ATT(g,t)$ by cohort and period	Heterogeneous effects and staggered adoption	Does not by itself correct compositional change in RCS data
DR-DiD	$ATT(g,t)$ with outcome regression and weighting	Observable covariates and double robustness	Depends on the available aggregate covariates
S&X approx.	2x2 cells by cohort and relative time, with four-category multinomial logit	Sensitivity to observable composition at the aggregate level	Does not replace the individual estimator with linked student-score data

Note: The last row is not the literal implementation of the individual-level estimator of Sant'Anna and Xu (2026); it is an aggregate approximation adapted to the absence of a reliable link between individual enrollment and individual scores.

effects generates event-study coefficients, interpreted as average effects by relative time to conversion.

The third specification incorporates covariates through the doubly robust estimator associated with Sant'Anna and Zhao (2020). The intuition is to combine two ways of constructing the counterfactual. The first is an outcome model, which adjusts for observable differences across schools. The second is a propensity model, which reweights control units to make them more similar to treated units. If at least one of these components is correctly specified, the estimator preserves desirable asymptotic properties. In this paper, the covariates include baseline proficiency, municipal controls, and aggregate composition measures constructed from enrollment microdata.

The fourth specification is the most important adaptation in this paper. Sant'Anna and Xu (2026) show that, in repeated cross-sections, usual DiD identification may fail when treatment changes the composition of observed individuals. Their original proposal models the distribution of individual covariates and constructs counterfactuals that accommodate compositional changes. In applications of this type, this involves estimating probabilities of belonging to groups and periods, for instance through multinomial logits, and using weighting or matching to compare similar individual distributions.

The data design in this paper prevents a literal implementation. I do not observe, in the same reliable record, individual enrollment and individual SARESP scores. Therefore, I cannot estimate the model at the student level or perform individual matching between treated and control students based on the same set of characteristics and outcomes. The adaptation shifts the problem to the school-year-grade level. It does not perfectly decompose learning and selection, but it tests whether incorporating observable composition changes the estimated effect.

For each cohort g and relative time k, I construct a 2x2 comparison using years $g-1$ and $g+k$. Define $D_i = 1$ for schools in cohort g and $T_i = 1$ for observations in period $g+k$. This generates four cells:

$$(D_i, T_i) \in \{(0,0), (0,1), (1,0), (1,1)\}. \tag{6}$$

Cell $(1,1)$ contains treated schools in the post-treatment period; $(1,0)$ contains the same schools in the period immediately before conversion; $(0,1)$ and $(0,0)$ contain control schools in the same years, respectively. The first step is to estimate a multinomial logit for the probability of each cell conditional on baseline covariates:

$$p_{dt}(X_i) = Pr(D_i = d, T_i = t \mid X_i), \quad d, t \in \{0,1\}. \tag{7}$$

Covariates X_i are fixed in year $g-1$, before cohort conversion. This choice avoids turning post-treatment characteristics into bad controls. In the implementation, I use the composition covariates with the best support in the aggregate panel: share of female students, average age, share of Black or mixed-race students, share of Bolsa Familia beneficiaries, and log enrollment. The variables are standardized within each 2x2 cell to improve the numerical stability of the multinomial logits.

From these probabilities, I construct Hajek weights for each cell. The operator $E_n[\cdot]$ denotes the sample mean and, in the denominator below, normalization is carried out within cell (d,t):

$$w_{dt}(D_i, T_i, X_i) = \frac{\mathbb{I}\{D_i = d, T_i = t\}\, p_{11}(X_i)/p_{dt}(X_i)}{E_n[p_{11}(X_i)/p_{dt}(X_i) \mid D_i = d, T_i = t]}. \tag{8}$$

These weights reweight the other cells to the observable distribution of the treated post-treatment cell, $(1,1)$. I then estimate three outcome regressions, one for each required counterfactual cell:

$$m_{dt}(X_i) = E[Y_i \mid D_i = d, T_i = t, X_i], \quad (d,t) \in \{(1,0),(0,1),(0,0)\}. \tag{9}$$

The aggregate estimator inspired by S&X combines weighting and outcome regression. For each pair (g,k), the estimated function is:

$$\widehat{\tau}^{SX}_{gk} = E_n\left[w_{11}(Y - m_{10} - m_{01} + m_{00}) - w_{10}(Y - m_{10}) - w_{01}(Y - m_{01}) + w_{00}(Y - m_{00})\right]. \tag{10}$$

The interpretation is direct: the first component uses the treated post-treatment cell, while the remaining components adjust the counterfactual evolution using the other three cells. The effects $\widehat{\tau}_{gk}^{SX}$ are then aggregated by relative time, with weights proportional to the number of treated schools observed in each post-treatment cell.

This form is the augmented version of the 2x2 difference-in-differences. If the regression terms were omitted, the estimator would collapse to the weighted difference $E_n[w_{11}Y] - E_n[w_{10}Y] - E_n[w_{01}Y] + E_n[w_{00}Y]$. The terms $m_{dt}(X_i)$ recenter each counterfactual cell through outcome regression, preserving the DiD intuition while adding adjustment for observable covariates.

This procedure required an important common-support restriction. Because treated schools account for a small share of the sample, some cells generate $p_{11}(X_i)$ probabilities very close to zero for a large share of controls. An initial attempt to use broader propensity weighting revealed this issue: excessively loose support thresholds allowed more cells to be estimated, but generated extreme weights and estimates dominated by regions without substantive comparison. The final version therefore does not force those cells. When the multinomial logit indicates insufficient common support, the cell is excluded from aggregation.

This choice is central for interpreting the results. The S&X approximation reported in this paper does not use a binary DR-DiD fallback and does not replace difficult cells with another estimator. It only reports cells estimated with a four-category multinomial logit and outcome regressions. In the final run, the procedure produces 26 valid cells for each 9th-grade subject and 24 valid cells for each high-school subject; the remaining exclusions are due to insufficient common support.

All event studies are reported with a universal baseline at $k = -1$, meaning that the year immediately before conversion is normalized to zero. This follows recent guidance for interpreting event studies using a common baseline (Baker et al., 2025; Roth, 2026). Standard errors are clustered at the school level whenever allowed by the estimator. For the aggregate S&X approximation, inference is built from the variances of the 2x2 comparisons and should be interpreted as approximate.

Finally, to evaluate the empirical relevance of compositional change, I compare DR-DiD with the S&X approximation through a Hausman-type diagnostic. The logic is simple: if the aggregate compositional correction systematically changes the estimator, the difference between the two should appear in the results. The interpretation, however, is necessarily cautious. Because the S&X approximation is estimated with few cells and the variance of the difference is constructed conservatively, high p-values should be read as absence of evidence of a statistical difference, not as proof that relevant compositional change is absent.

4.4 Threats to Identification

The empirical strategy above approximates a counterfactual for PEI schools using not-yet-treated or never-treated schools. This comparison is more appropriate than a simple difference in means, but it does not eliminate all alternative explanations. Four threats are especially relevant in this paper.

The first is mean reversion. Because some schools that joined PEI had lower performance before conversion, part of the subsequent growth could have occurred even without the program. In simple terms, schools with very low scores in one year tend to experience some mechanical recovery in following years even if no new policy is implemented. To assess this possibility, Section 5 presents a diagnostic that restricts the sample to the baseline-proficiency support of treated schools and compares the evolution of never-treated controls with low and high initial performance.

The second threat is spillovers. The literature on PEI documents that regular schools near converted schools may be affected by student migration and changes in local composition. If part of the control group is negatively affected by PEI expansion, the estimated counterfactual lies below what would have occurred in the absence of the program, and the ATT may be overestimated. As a diagnostic, I re-estimate DR-DiD excluding controls located in school districts or school-district-by-neighborhood pairs that already had at least one active PEI school in the year. This strategy does not replace a measure of geographic distance between schools, but it tests whether the main magnitude depends on administratively nearby controls.

The third threat comes from missing data in composition covariates. The main specification uses annual-mean imputation to preserve panel coverage. This avoids making the sample depend only on schools with more complete administrative records, but it also introduces an additional assumption: imputed information should not create artificial differences between treated and control schools. The no-imputation robustness exercise, presented in Section 5, shows that requiring complete covariates sharply reduces the sample and changes the estimated magnitudes. For this reason, this exercise is read as a sensitivity diagnostic, not as a direct replacement for the main specification.

The fourth threat is selection into SARESP participation. Even if enrollment remains observable, the set of students who actually take the exam may change after conversion to PEI. If lower-proficiency students stop taking the test at higher rates, observed average proficiency may increase without fully reflecting learning. Section 5 presents a diagnostic of the ratio between SARESP participants and enrollment, which recommends additional caution for high-school results.

These threats do not automatically invalidate the empirical design, but they delimit its interpretation. The paper estimates average gains associated with PEI adoption in an aggregate school panel. It does not prove, in isolation, that the entire observed gain comes exclusively from individual learning caused by the program. The contribution is to show that the gains remain positive under modern estimators for staggered adoption and under an aggregate compositional approximation, while also making explicit the channels that the available data still cannot fully close.

5 Results and Discussion

This section presents the empirical results. Section 5.1 documents descriptive statistics. Section 5.2 presents the estimated effects on proficiency, comparing TWFE, Callaway and Sant'Anna (2021), Sant'Anna and Zhao (2020), and the aggregate approximation inspired by Sant'Anna and Xu (2026). Section 5.3 discusses effect dynamics and the compositional diagnostic. Finally, Section 5.4 presents robustness checks.

5.1 Descriptive Statistics

The descriptive analysis relies on two complementary datasets. The first is the school-level aggregate panel used in the main estimation. The second consists of student microdata, used to construct composition covariates and document changes in the profile of enrolled students. For comparison, observations are classified into three groups: *Control*, schools that do not join PEI within the analysis window; *Treated Pre*, PEI schools before conversion; and *Treated Post*, PEI schools after conversion.

Table 2 presents statistics from the microdata. The column *Diff (Post–Pre)* reports the coefficient from a linear regression of the variable of interest on a post-treatment indicator, restricted to the treated group, with standard errors clustered at the school level.

Table 2. Descriptive Statistics – Microdata (Student Level)

	Group Mean			Compositional Test	
Variable	Control	Treated Pre	Treated Post	Diff (Post–Pre)	*p*-value
Share Female	0.501	0.502	0.528	+0.026	<0.001
Share Special Needs	0.006	0.006	0.013	+0.006	<0.001
Portuguese Accuracy (%)	54.83	54.19	66.18	+11.99	<0.001
Math Accuracy (%)	49.25	45.61	56.66	+11.05	<0.001

Note: Standard errors are clustered at the school level. Special needs is an indicator for at least one disability or condition recorded in the administrative registry.

The results in Table 2 show that conversion to PEI coincides with changes in student composition. The share of female students increases by 2.6 percentage points after conversion. The share of students with a registered special need also increases. This second result requires caution: it may reflect a real compositional change, but it may also reflect improved administrative recording in converted schools. For identification, the central point is that observed composition changes after treatment. Even when the substantive direction of the change is not trivial, the existence of the change matters.

Accuracy rates in Portuguese Language and Mathematics also increase sharply in the post-treatment period. These raw jumps, however, are not causal effects. They combine learning, student selection, changes in test participation, and pre-existing differences across schools. The task of the estimators presented below is to separate, as much as possible, the component attributable to the policy from the component associated with composition.

Table 3. Pre-Treatment Balance – Microdata

Variable	Control	Treated Pre	Diff	*p*-value
Share Female	0.501	0.502	+0.001	0.487
Share Special Needs	0.006	0.006	+0.000	0.072
Portuguese Accuracy (%)	54.83	54.19	−0.644	$<$0.001
Math Accuracy (%)	49.25	45.61	−3.641	$<$0.001

Note: Comparison restricted to the pre-treatment period. Standard errors are clustered at the school level.

Table 3 compares treated and control schools before adoption. Observed demographic dimensions are similar across groups, but schools that would later join PEI had lower pre-treatment performance, especially in Mathematics. This reinforces that the program was not randomly assigned across equivalent schools. Econometrically, the evidence justifies using estimators that control for fixed differences, allow staggered adoption, and incorporate baseline covariates.

Table 4 adds a direct diagnostic on exam participation. In 9th grade, the participation rate is quite similar before and after conversion, with a descriptive increase of 1.2 percentage points in Portuguese Language and 0.8 percentage points in Mathematics. In high school, however, the rate falls by around 16 percentage points after PEI adoption. This result does not automatically invalidate the proficiency estimates, but it opens a substantive interpretation channel: part of the estimated average gain in high school may reflect a change in the set of students actually observed in SARESP, in addition to learning or enrollment selection. For this reason, high-school results should be read with additional caution.

Table 4. Diagnostic of the SARESP participation rate

Grade-subject	Control	Treated pre	Treated post	Post–pre diff.
9th LP	84.3%	81.0%	82.2%	+1.2 p.p.
9th Mat	85.0%	81.6%	82.4%	+0.8 p.p.
HS3 LP	78.0%	71.5%	55.5%	-16.0 p.p.
HS3 Mat	78.8%	72.1%	55.8%	-16.3 p.p.

Note: The rate divides the number of students with recorded exam participation by total active state enrollments in the school-grade-year, using the harmonized denominator from the final panel. The post–pre difference is descriptive and restricted to schools that join PEI between 2012 and 2018.

Figure 5 shows the distribution of standardized proficiency in the post-treatment period. In every cell, the distribution of schools with active PEI is shifted to the right relative to the control or not-yet-treated group. This figure does not replace causal estimation, because it mixes program effects and selection of treated schools. Still, it is informative: the result the estimators seek to explain is not merely a small mean difference, but a visible separation in the proficiency distribution of schools exposed to PEI.

In the aggregate panel, average proficiency in PEI schools increases by approximately 29 points in Portuguese Language and 30 points in Mathematics after conversion. The movement is large on the SARESP scale, but it still cannot be interpreted as causal. It shows that there is a relevant aggregate gain to be explained; the empirical question is how much of this gain remains when the comparison across schools is corrected.

5.2 Main Results

Table 6 presents the average post-treatment effect for each grade-subject combination. The reported number corresponds to the mean of coefficients from $k = 0$ to $k = 4$, with period

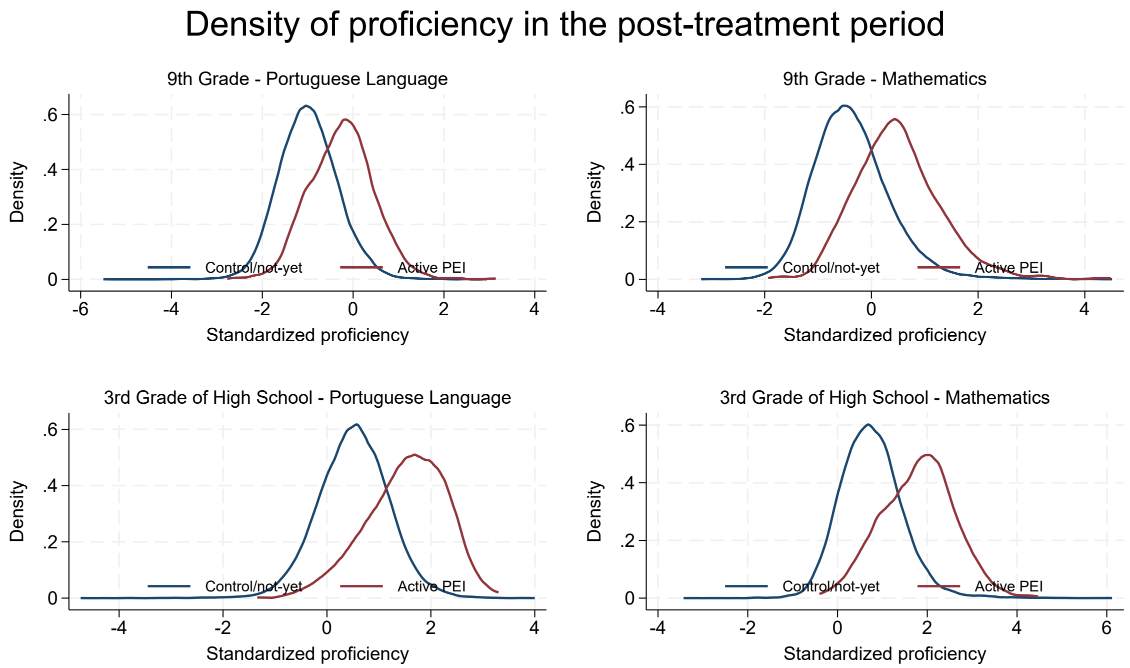


Figure 5. Density of proficiency in the post-treatment period

Source: Author's elaboration. Treatment denotes school-year observations with active PEI; control denotes not-yet-treated or never-treated schools in the same calendar year.

Table 5. Descriptive Statistics – Aggregate Panel (School Level)

	Group Mean			Diff (Post–Pre)	
Variable	Control	Treated Pre	Treated Post	Coef.	*p*-value
Mean Proficiency, Portuguese	243.8	244.7	273.5	+28.8	<0.001
Mean Proficiency, Math	254.0	254.7	284.9	+30.2	<0.001
Share Urban	0.959	1.000	1.000	—	—
Share Settlement	0.004	0.000	0.000	—	—

Note: Proficiency is reported in the SARESP scale. "—" indicates no variation in the treated group for the respective characteristic.

$k = -1$ normalized as the common baseline.

Table 6. Average post-treatment effect (standard deviations)

Grade-subject	CS21	DR-DiD	S&X approx.	TWFE
9th LP	0.491***	0.510***	0.502***	0.475***
	(0.019)	(0.019)	(0.028)	(0.017)
9th Mat	0.585***	0.601***	0.603***	0.583***
	(0.020)	(0.020)	(0.033)	(0.018)
HS3 LP	0.526***	0.571***	0.601***	0.507***
	(0.022)	(0.022)	(0.035)	(0.019)
HS3 Mat	0.605***	0.627***	0.675***	0.604***
	(0.022)	(0.021)	(0.037)	(0.019)

Note: average from $k = 0$ to $k = 4$ of the event study with universal baseline at $k = -1$, estimated in the 2011–2018 school-year-grade-subject panel. Conservative standard errors in parentheses, computed from the variances of period-specific coefficients; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The full results by relative period, including standard errors for each estimator and grade-subject cell, are reported in the appendix of the original monograph. This organization keeps the main text focused on the summary of average effects while preserving the complete estimation paths for inspection.

Effects are positive in all cells and for all estimators. In 9th grade, DR-DiD estimates an average effect of 0.510 standard deviations in Portuguese Language and 0.601 in Mathematics. The S&X approximation produces very similar results: 0.502 and 0.603 standard deviations, respectively. In high school, effects are even larger. DR-DiD estimates 0.571 standard deviations in Portuguese Language and 0.627 in Mathematics; the S&X approximation estimates 0.601 and 0.675.

The magnitude is substantive. Since the average annual standard deviation of proficiency in the panel is on the order of several dozen SARESP points, effects between approximately 0.50 and 0.68 standard deviations represent large shifts within the distribution of school performance. Reading the results in standard deviations should therefore not be seen as a statistical abstraction: it corresponds to relevant differences on the concrete exam scale.

These values are also high relative to the literature. Using linked data and focusing on Sao Paulo high schools, Scorzafave et al. (2023) estimate gains of 0.31 standard deviations in Mathematics and 0.21 in Portuguese Language. In Pernambuco, reported magnitudes for full-time schooling are around 0.20 standard deviations (Rosa et al., 2022). This difference should not automatically be interpreted as evidence that the pedagogical effect estimated here is larger. This paper uses an aggregate school panel, annual standardization, its own window and sample, and does not directly observe the student-score link. For this reason, the main magnitudes should be read as aggregate evidence of gains associated with PEI and as potentially sensitive to the unit of analysis, SARESP participation, control-group composition, and covariate coverage.

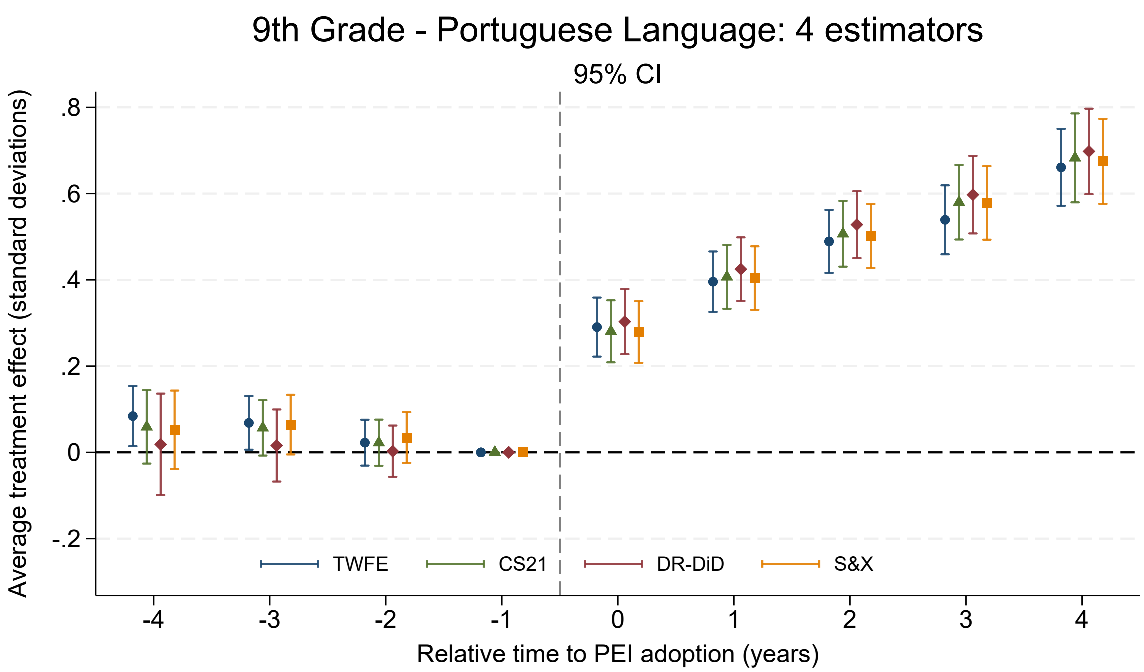
The central point of the table is the relative stability of the estimates. Moving from CS21 to DR-DiD, which incorporates baseline proficiency, municipal controls, and aggregate composition covariates, does not eliminate the effect. Incorporating the S&X approximation, now estimated only with four-category multinomial logits and without a binary fallback, also does not reduce the magnitude of the results. In high school, the S&X estimate is even larger than the DR-DiD estimate, especially in Mathematics. Conservative variances and the small number of cells prevent precise inference about this difference; the available evidence is consistent with stability between DR-DiD and S&X, but it cannot rule out smaller compositional effects.

TWFE appears with similar magnitudes, but should be read only as a reference. The policy has staggered adoption and effects grow with time since conversion. In this context, TWFE may combine inappropriate comparisons between already-treated schools and schools treated later. The main interpretation of the paper relies on CS21, DR-DiD, and the S&X approximation, always conditional on the identification assumptions discussed in Section 4.

5.3 Effect Dynamics

Figures 6–9 show the trajectory of event-study coefficients. In general, effects appear in the year of conversion and grow in subsequent years. This pattern could reflect dynamic student selection, changes in exam participation, or recovery among schools initially below the mean; it is also compatible with a policy that takes time to reorganize pedagogical practices, school management, teacher dedication, and the use of the extended school day. Because these channels cannot be separated with the available data, the dynamics should be read as temporal evidence of the estimated aggregate effect, not as a direct test of mechanisms.

Figure 6. 9th Grade - Portuguese Language

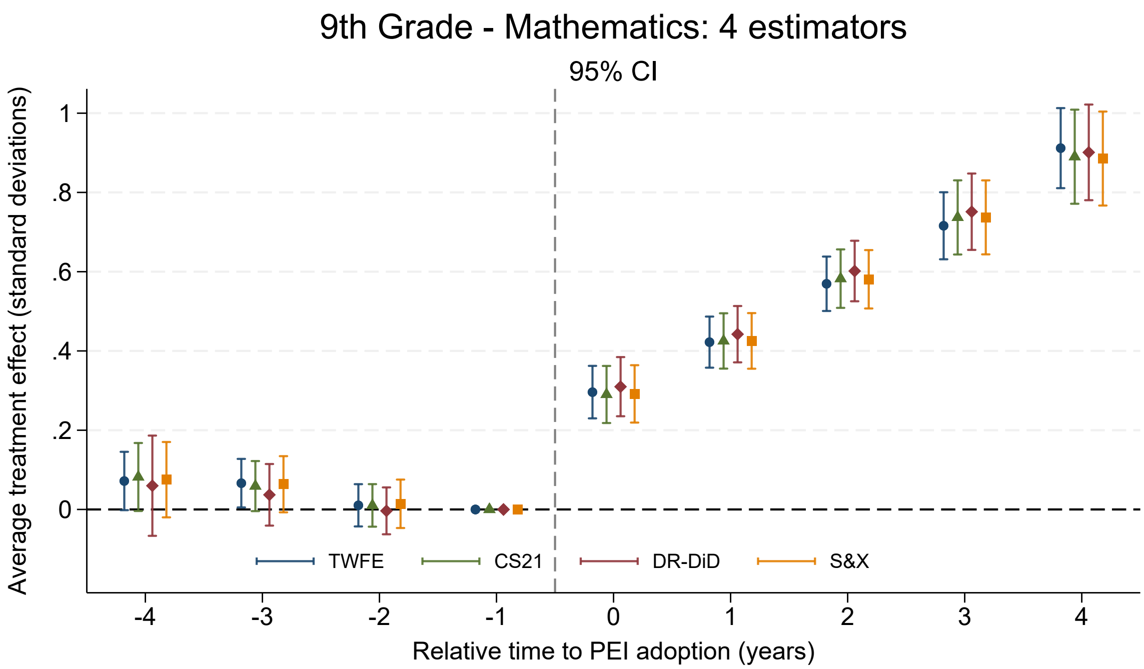


Source: author's elaboration. ATT coefficients by relative time to conversion; $k = 0$ is the year of PEI adoption and $k = -1$ is the normalized baseline period. Bands indicate 95% confidence intervals.

Pre-treatment coefficients are smaller than post-treatment effects and, in the preferred specifications, do not form a systematic anticipation pattern. There are isolated differences in some cells, which is expected in an exercise with multiple grades, subjects, and estimators. Still, the main dynamic is clear: estimated post-treatment effects are positive, persistent, and increasing. This reading does not eliminate concern about mean reversion: pre-treatment balance shows that schools that would later join PEI had lower performance, especially in Mathematics, so part of the observed growth could reflect mechanical convergence. The baseline-support diagnostic below reduces, but does not eliminate, this threat.

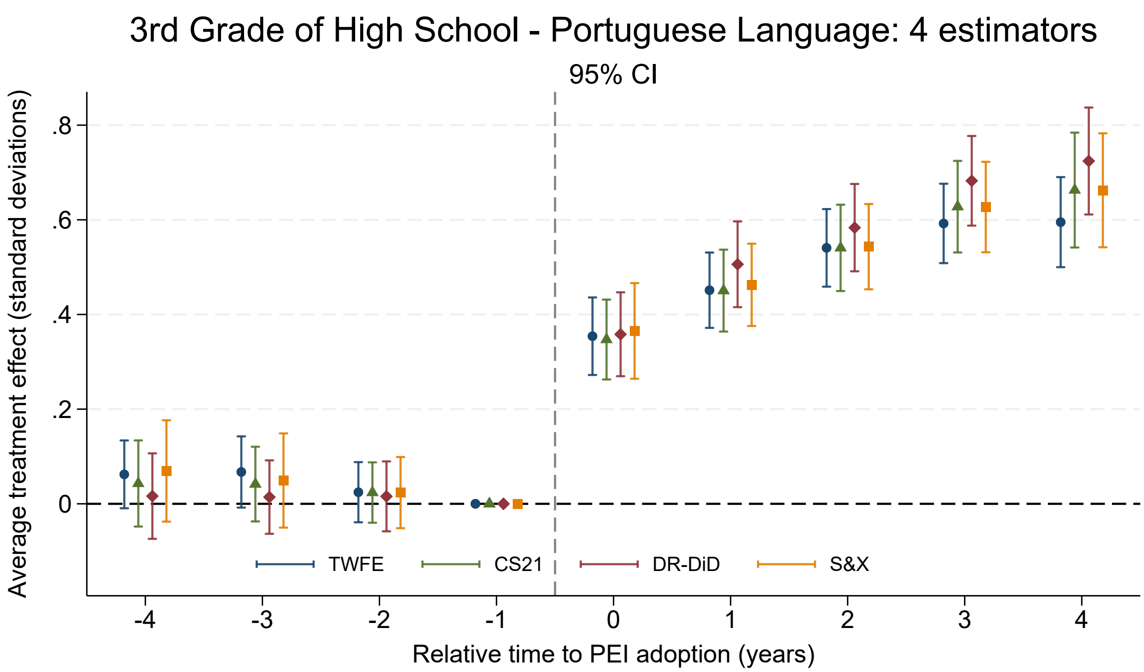
The dynamics suggest stronger effects in Mathematics than in Portuguese Language.

Figure 7. 9th Grade - Mathematics



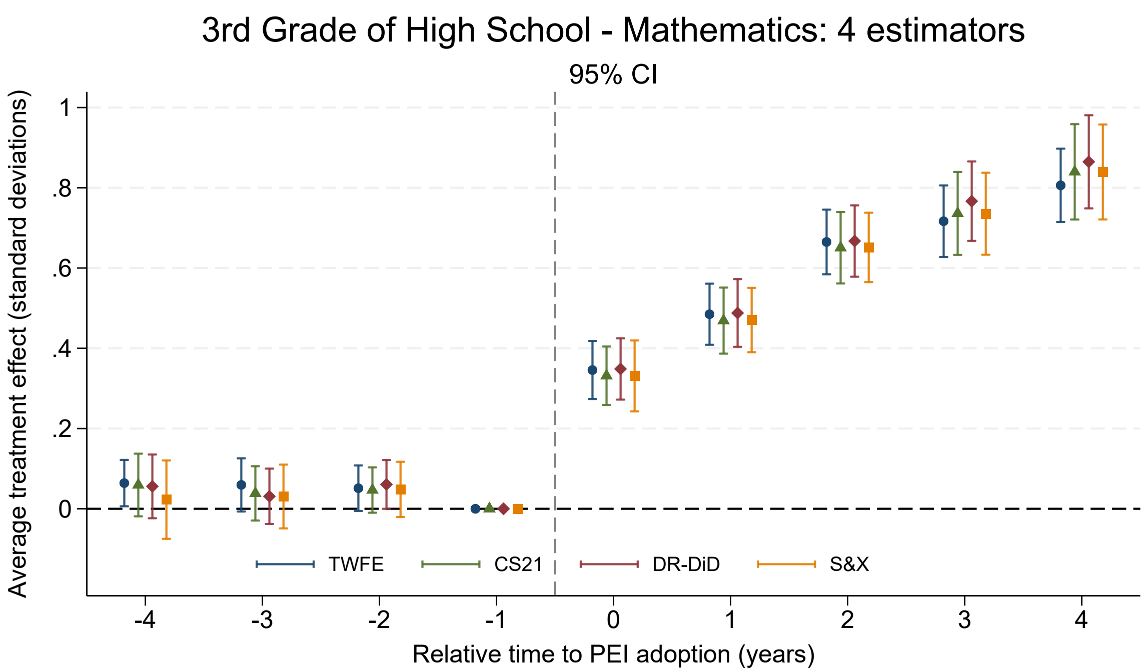
Source: author's elaboration. ATT coefficients by relative time to conversion; $k = 0$ is the year of PEI adoption and $k = -1$ is the normalized baseline period. Bands indicate 95% confidence intervals.

Figure 8. 3rd Grade of High School - Portuguese Language



Source: author's elaboration. ATT coefficients by relative time to conversion; $k = 0$ is the year of PEI adoption and $k = -1$ is the normalized baseline period. Bands indicate 95% confidence intervals.

Figure 9. 3rd Grade of High School - Mathematics



Source: author's elaboration. ATT coefficients by relative time to conversion; $k = 0$ is the year of PEI adoption and $k = -1$ is the normalized baseline period. Bands indicate 95% confidence intervals.

This difference should be interpreted as descriptive heterogeneity by subject, not as a definitive test of mechanisms. The available data do not allow me to distinguish whether the difference comes from learning, student selection, exam participation, or a combination of these channels.

5.4 Composition and Robustness

The central concern of the paper is that part of the observed gain in PEI schools may be generated by compositional change. Before looking at p-values, it is important to qualify the diagnostic: the comparison between DR-DiD and the S&X approximation uses only 24 to 26 valid cells by grade-subject and adopts a conservative variance equal to the sum of marginal variances. Therefore, the test has low power to detect small or moderate differences. Table 7 presents this comparison and also reports the minimum detectable effect with 80% power.

Table 7. Hausman-type diagnostic: DR-DiD versus S&X approximation

Grade-subject	DR-DiD	S&X approx.	Diff.	p-value	MDE 80%
9th grade – LP	0.510	0.502	-0.008	0.9630	0.198
9th grade – Mathematics	0.601	0.603	+0.002	0.9976	0.219
HS3 – LP	0.571	0.601	+0.031	0.9000	0.248
HS3 – Mathematics	0.627	0.675	+0.048	0.9057	0.243

Note: DR-DiD and S&X values correspond to the average post-treatment effect, computed as the mean from $k = 0$ to $k = 4$. The test uses a conservative variance equal to the sum of marginal variances. The 80% MDE is the minimum detectable difference, in standard deviations, with 80% power and a two-sided 5% test. The S&X approximation excludes cells without sufficient common support and does not use a binary DR-DiD fallback.

The p-values are high in all cells, but the minimum detectable difference ranges from 0.198 to 0.248 standard deviations. The correct reading is therefore restricted: given the available aggregate covariates and the conservative variance used in the test, there is no evidence that the compositional approximation produces statistically different effects from DR-DiD, but the test does not rule out differences smaller than approximately 0.20–0.25 standard deviations. Nor does it prove the absence of student selection, or replace an individual-level estimator with perfectly linked microdata. The p-values should be read as a sensitivity diagnostic, not as strong confirmation of compositional robustness.

It is also important to note what the S&X approximation was actually able to estimate. In 9th grade, there were 26 valid cells per subject. In high school, there were 24. Excluded cells were not replaced by another estimator: they were removed because of insufficient common support in the multinomial logit. This point matters because it avoids reporting as S&X an estimate that, in practice, would merely be binary DR-DiD in a subsample.

Table 8 presents a diagnostic for mean reversion. The baseline-support column re-estimates DR-DiD restricting the sample to the interval between the 10th and 90th percentiles of baseline proficiency among treated schools. This restriction forces the comparison to occur in a region of the distribution closer to the schools that joined PEI.

Table 8. Mean-reversion diagnostic

Grade-subject	Main DR-DiD	Baseline support	Δ low controls	Δ high controls
9th LP	0.510	0.533	0.239	-0.182
9th Mat	0.601	0.612	-0.172	-0.482
HS3 LP	0.571	0.581	-0.094	-0.578
HS3 Mat	0.627	0.607	-0.295	-0.542

Note: The baseline-support column re-estimates DR-DiD restricting the sample to the p10–p90 interval of baseline proficiency among treated schools. The last two columns show the average change, in standard deviations, between 2011 and 2018 for never-treated controls below/above the treated-school baseline median.

The results change little relative to the main specification. In 9th grade, the Portuguese Language effect moves from 0.510 to 0.533 standard deviations and the Mathematics effect from 0.601 to 0.612. In high school, Portuguese Language moves from 0.571 to 0.581, while Mathematics moves from 0.627 to 0.607. This exercise does not fully eliminate the possibility of mean reversion, but it reduces the concern that the results are driven only by comparisons between initially weak treated schools and very distant controls at baseline. The last two columns also show that never-treated controls with low initial performance do not display a mechanical recovery capable of reproducing the magnitude of the estimated PEI effects.

Table 9 evaluates the sensitivity of the results to possible regional contamination of the control group. The first restriction removes controls in school districts that already had at least one active PEI school in the year; the second uses a finer unit, the school-district-by-neighborhood pair.

Table 9. Regional spillover diagnostic

Grade-subject	Restriction	Main DR-DiD	Restricted DR-DiD	Restricted N
9th LP	DE	0.510	0.482	13.739
9th LP	DISTR	0.510	0.526	21.097
9th Mat	DE	0.601	0.577	13.743
9th Mat	DISTR	0.601	0.615	21.107
HS3 LP	DE	0.571	0.563	13.221
HS3 LP	DISTR	0.571	0.571	20.947
HS3 Mat	DE	0.627	0.654	13.228
HS3 Mat	DISTR	0.627	0.634	20.959

Note: the restriction removes controls located in school districts (DE) or DE-neighborhood pairs (DISTR) that already had at least one active PEI school in the year. The estimator is DR-DiD; standard errors are omitted to keep the table compact. This is a regional proxy for spillover contamination of the counterfactual, not a measure of geographic distance.

The estimates remain positive and close to the main estimates. Excluding controls by school district moderately reduces effects in 9th grade, from 0.510 to 0.482 in Portuguese Language and from 0.601 to 0.577 in Mathematics, but does not change the qualitative conclusion. In high school, effects also remain close. The school-district-by-neighborhood restriction is even more stable. Because this approach uses administrative proximity rather than geographic distance between schools, it should be read as a partial spillover diagnostic, not as a definitive solution for this channel. School districts may cover broad areas, while the mechanism documented by Santos (2022) operates through effective proximity between schools and student migration.

Table 10 presents a sensitivity diagnostic to imputation, repeating DR-DiD with complete raw covariates and no annual-mean imputation.

Table 10. Sensitivity to covariate imputation

Grade-subject	DR-DiD (main)	DR-DiD no imputation	Main N	No-imputation N	N fraction
9th LP	0.510	0.384	24754	10804	0.436
9th Mat	0.601	0.502	24769	10819	0.437
HS3 LP	0.571	–	27515	–	–
HS3 Mat	0.627	–	27530	–	–

Note: the no-imputation version uses only observations with complete raw covariates. The exercise should be read as a diagnostic of sensitivity and sample coverage, not as a direct substitute for the main specification.

The diagnostic reveals an important tension. Requiring complete covariates reduces the 9th-grade sample to about 44% of the main sample and makes estimation infeasible in high school. In the two cells where estimation runs, effects remain positive, but fall to 0.384 standard deviations in Portuguese Language and 0.502 in Mathematics. This exercise does not replace the main specification, because it changes the sample composition; its function is to show that missing covariate data are empirically relevant and that annual-mean imputation preserves broader coverage of the state network.

Table 11 helps interpret this decline. In 9th grade, complete and imputed observations have very similar baseline proficiency, but the complete sample contains a smaller share of treated schools. In high school, the contrast is more severe: there are just over one thousand complete observations against almost 28 thousand with at least one imputed covariate, and the treated share in the complete sample is about one third of the share observed among imputed observations. Therefore, the no-imputation specification is better interpreted as a diagnostic of sensitivity and data coverage, not as a direct substitute for the main specification. In high school, the no-imputation column is empty because the loss of coverage prevents comparable estimation.

Taken together, the results indicate persistent gains associated with PEI adoption at the

Table 11. Diagnostic of imputed covariates

Grade-subject	Complete N	Imputed N	Complete baseline	Imputed baseline
9th LP	13.992	15.487	230.07	229,75
9th Mat	13.999	15.495	246,56	245,87
HS3 LP	1.032	27.842	266,68	265,49
HS3 Mat	1.033	27.857	272,48	269,72

Note: complete observations have all raw covariates used in the no-imputation DR-DiD. Baseline differences across columns indicate that the no-imputation diagnostic also changes sample composition. Baseline is mean proficiency on the SARESP scale.

aggregate level, although the magnitude should be interpreted cautiously given the sensitivity to the no-imputation sample, the decline in SARESP participation in high school, and the limitations of the compositional test. The evidence is stronger for the question the data can answer: schools converted to the full-time model experience average proficiency gains relative to comparable schools within the 2011–2018 window. The evidence is more limited for the question that would require fully integrated microdata: how much of this gain comes exclusively from individual learning, net of all selection and changes in exam participation. This distinction is the empirical frontier of the paper. The exercise shows that observable aggregate composition does not eliminate the results, but it also makes clear that a complete individual decomposition remains beyond the reach of the available data.

6 Conclusion

This paper evaluated proficiency gains associated with adoption of the Full-Time Education Program in Sao Paulo state schools using SARESP data from 2011 to 2018. The central question was not only whether PEI schools improved after conversion, but whether this gain remains once the empirical design is taken seriously: staggered adoption, heterogeneous effects over time, observable differences across schools, and compositional change in the student body. The main estimates indicate average post-treatment effects between approximately 0.50 and 0.68 standard deviations, with larger effects in Mathematics and in high school. The increasing dynamics in the event studies could reflect dynamic student selection, changes in exam participation, or recovery among schools initially below the mean; they are also compatible with pedagogical and organizational maturation of the program. The available data do not allow these channels to be fully separated.

The substantive contribution of the paper is to show that these results do not depend only on a simple comparison between treated schools and controls. The gains remain when TWFE is abandoned as the main estimator, when Callaway and Sant'Anna (2021) is used to handle staggered adoption, when covariates are added through Sant'Anna and Zhao (2020), and when an aggregate approximation inspired by Sant'Anna and Xu (2026) is implemented. This last step is important because the literature on PEI shows that school conversion changes student composition. Therefore, an evaluation of the program that ignores this channel risks confusing learning gains with student selection. Within the observable dimensions available in this paper, the compositional correction does not eliminate the estimated effects, but the Hausman-type diagnostic has low power and is not evidence against compositional effects of up to approximately 0.20–0.25 standard deviations.

The methodological contribution is to make explicit the distance between the ideal estimator and the feasible estimator. The estimator of Sant'Anna and Xu (2026) was designed for repeated cross-sections in which compositional change can be handled at the individual level. To apply it fully, one would need to observe, in the same record, individual enrollment, student covariates, and SARESP scores. This structure is not publicly available for the period analyzed. For this reason, this paper builds an aggregate adaptation: for each cohort and relative time, I estimate 2x2 cells using a four-category multinomial logit, composition weights, and outcome regressions, always using covariates fixed at the cohort baseline. The final version does not use a binary S&Z fallback; when common support is insufficient, the cell is excluded. This makes the estimate more restrictive, but also more honest.

This limitation defines the scope of interpretation. The results are evidence of aggregate proficiency gains in PEI schools; they are not a perfect decomposition between individual learning, student selection, and changes in exam participation. Even so, the exercise is informative. If observable selection were the main explanation for the gains, one would expect the estimates to fall substantially when moving from CS21 to DR-DiD and to the S&X approximation. This is not what occurs. In addition, the diagnostics for mean reversion and regional spillovers preserve magnitudes close to the main estimates. On the other hand, the no-imputation sensitivity check shows that the magnitude depends on coverage of enrollment covariates, especially because the complete-case sample is much smaller and, in high school, insufficient to sustain the same estimation. The participation diagnostic for SARESP also recommends additional caution in high school, where the ratio between participants and enrollment falls after conversion.

The main open questions follow from this same data restriction. The first natural extension is to obtain access to administrative databases that make it possible to link individual enrollment, exam participation, and individual scores, approximating the Sant'Anna and Xu estimator more faithfully and separating learning, selection, and exam attendance more clearly. The second is to incorporate more direct geographic measures of exposure to nearby PEI schools, using distance or commuting time, to move beyond the proxy based on school district and neighborhood. The third is to extend the window beyond 2018, separately treating the pandemic, changes in assessment, and the accelerated expansion of PEI. In sum, this paper shows that PEI is associated with persistent proficiency gains in an aggregate school panel. The conclusion does not settle the discussion about mechanisms, but it organizes its terms more clearly: there is a positive aggregate gain, there is compositional change, there is evidence of altered high-school participation, and the observable part of composition does not appear sufficient to overturn the results under the estimated specifications. At the same time, the magnitudes obtained here are larger than those estimated by Scorzafave et al. (2023) for the same program, reinforcing that the prudent interpretation is to treat the results as aggregate evidence conditional on the available design, not as a definitive measure of the individual net effect of PEI.

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A SARESP Score Intervals

The SARESP score interval is the rule used to classify a student's proficiency level given her score on the exam. The data for the table are available on the SARESP page of the Education Open Data portal. The grades used in the empirical analysis, 9th grade and the 3rd grade of high school, are highlighted in bold.

Table 12. Score Intervals

Portuguese Language

	3rd	5th	7th	**9th**	**HS3**
Below Basic	<125	<150	<175	**<200**	**<250**
Basic	125 to 174	150 to 199	175 to 224	**200 to 274**	**250 to 299**
Adequate	175 to 254	200 to 249	225 to 274	**275 to 324**	**300 to 374**
Advanced	≥225	≥250	≥275	**≥325**	**≥375**

Mathematics

	3rd	5th	7th	**9th**	**HS3**
Below Basic	<150	<175	<200	**<225**	**<275**
Basic	150 to 199	175 to 224	200 to 249	**225 to 299**	**275 to 349**
Adequate	200 to 249	225 to 274	250 to 299	**300 to 349**	**350 to 399**
Advanced	≥250	≥275	≥300	**≥350**	**≥400**

Source: Education Open Data, author's elaboration.

B Complete Main-Model Results

This appendix presents the complete tabular results for the main specification. The corresponding figures are shown in the main text, in Figures 6–9.

B.1 ATT by Period

Table 13. Main model, 9th grade, Portuguese Language: ATT by relative period

Relative period	TWFE	CS21	DR-DiD	S&X approx.
-4	0.084 (0.036)	0.059 (0.043)	0.019 (0.060)	0.047 (0.050)
-3	0.068 (0.032)	0.057 (0.033)	0.016 (0.043)	0.080 (0.037)
-2	0.022 (0.027)	0.022 (0.027)	0.003 (0.030)	0.044 (0.031)
-1	0.000	0.000	0.000	0.000
0	0.290 (0.035)	0.281 (0.037)	0.303 (0.039)	0.292 (0.045)
1	0.396 (0.036)	0.407 (0.038)	0.425 (0.038)	0.416 (0.050)
2	0.489 (0.037)	0.507 (0.039)	0.528 (0.040)	0.509 (0.056)
3	0.539 (0.041)	0.580 (0.044)	0.598 (0.046)	0.597 (0.067)
4	0.661 (0.046)	0.683 (0.053)	0.698 (0.051)	0.695 (0.087)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 14. Main model, 9th grade, Mathematics: ATT by relative period

Relative period	TWFE	CS21	DR-DiD	S&X approx.
-4	0.072 (0.038)	0.082 (0.044)	0.060 (0.065)	0.076 (0.053)
-3	0.066 (0.031)	0.059 (0.032)	0.037 (0.040)	0.079 (0.038)
-2	0.010 (0.027)	0.010 (0.027)	-0.003 (0.030)	0.025 (0.032)
-1	0.000	0.000	0.000	0.000
0	0.296 (0.034)	0.290 (0.037)	0.310 (0.038)	0.311 (0.046)
1	0.422 (0.033)	0.425 (0.036)	0.442 (0.036)	0.442 (0.050)
2	0.569 (0.035)	0.583 (0.038)	0.602 (0.039)	0.595 (0.060)
3	0.716 (0.043)	0.737 (0.048)	0.751 (0.049)	0.762 (0.080)
4	0.912 (0.052)	0.890 (0.061)	0.901 (0.062)	0.907 (0.109)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 15. Main model, HS3, Portuguese Language: ATT by relative period

Relative period	TWFE	CS21	DR-DiD	S&X approx.
-4	0.062 (0.037)	0.043 (0.046)	0.016 (0.046)	0.044 (0.089)
-3	0.067 (0.038)	0.042 (0.040)	0.014 (0.040)	0.112 (0.063)
-2	0.025 (0.032)	0.024 (0.033)	0.016 (0.038)	0.043 (0.045)
-1	0.000	0.000	0.000	0.000
0	0.354 (0.042)	0.347 (0.043)	0.358 (0.045)	0.434 (0.071)
1	0.451 (0.041)	0.450 (0.044)	0.506 (0.046)	0.541 (0.070)
2	0.541 (0.042)	0.541 (0.047)	0.583 (0.047)	0.645 (0.076)
3	0.592 (0.043)	0.628 (0.049)	0.682 (0.048)	0.665 (0.087)
4	0.595 (0.049)	0.663 (0.062)	0.724 (0.058)	0.722 (0.090)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 16. Main model, HS3, Mathematics: ATT by relative period

Relative period	TWFE	CS21	DR-DiD	S&X approx.
-4	0.064 (0.029)	0.059 (0.040)	0.056 (0.041)	0.075 (0.077)
-3	0.059 (0.034)	0.038 (0.035)	0.031 (0.035)	0.091 (0.049)
-2	0.051 (0.029)	0.047 (0.029)	0.061 (0.031)	0.087 (0.041)
-1	0.000	0.000	0.000	0.000
0	0.346 (0.037)	0.332 (0.037)	0.349 (0.039)	0.410 (0.065)
1	0.485 (0.039)	0.469 (0.042)	0.488 (0.043)	0.552 (0.067)
2	0.665 (0.041)	0.651 (0.045)	0.667 (0.045)	0.750 (0.081)
3	0.717 (0.046)	0.736 (0.053)	0.767 (0.051)	0.776 (0.094)
4	0.806 (0.047)	0.840 (0.061)	0.865 (0.059)	0.887 (0.103)

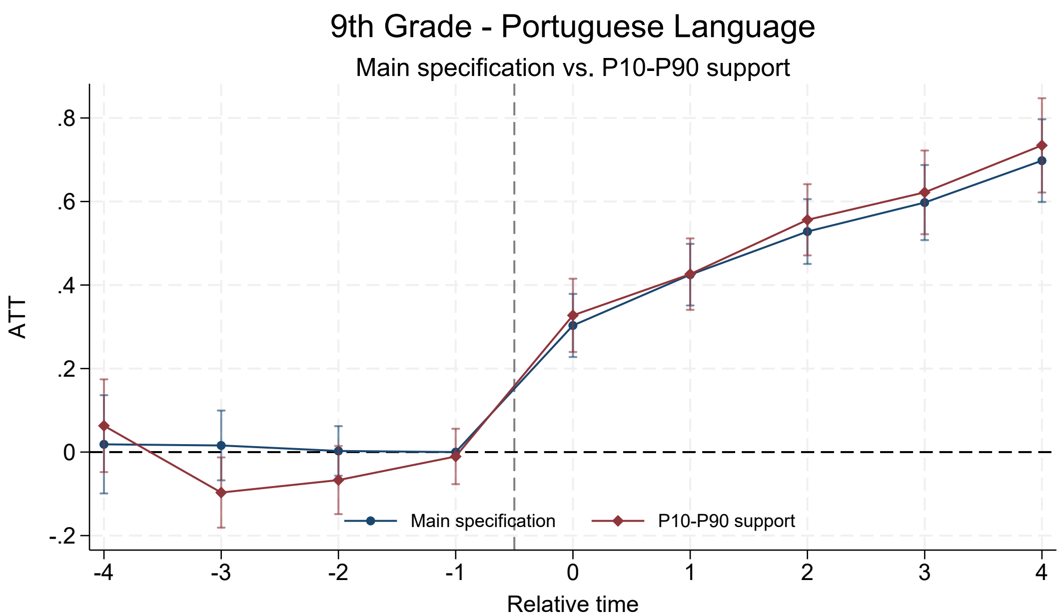
Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

C Mean-Reversion Diagnostic

This appendix compares the main specification with the baseline-support restriction. The goal is to assess whether the main results depend on comparing initially low-performing treated schools with controls far away in the pre-treatment distribution.

C.1 Event Studies by Model

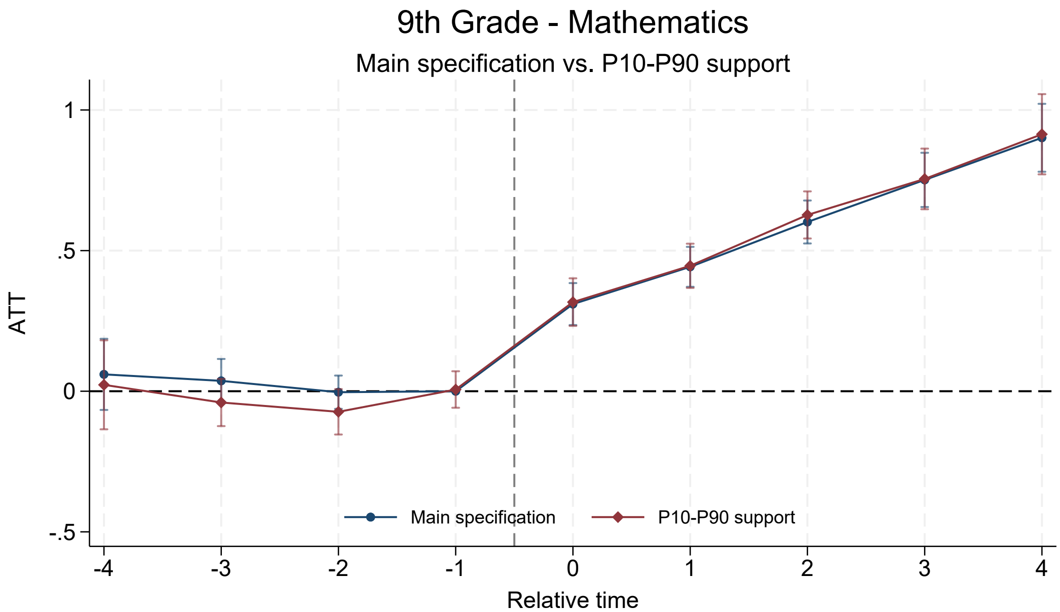
Figure 10. Mean-reversion diagnostic: 9th grade, Portuguese Language



Source: author's elaboration. The restricted line uses schools within the P10–P90 support of baseline proficiency among treated schools.

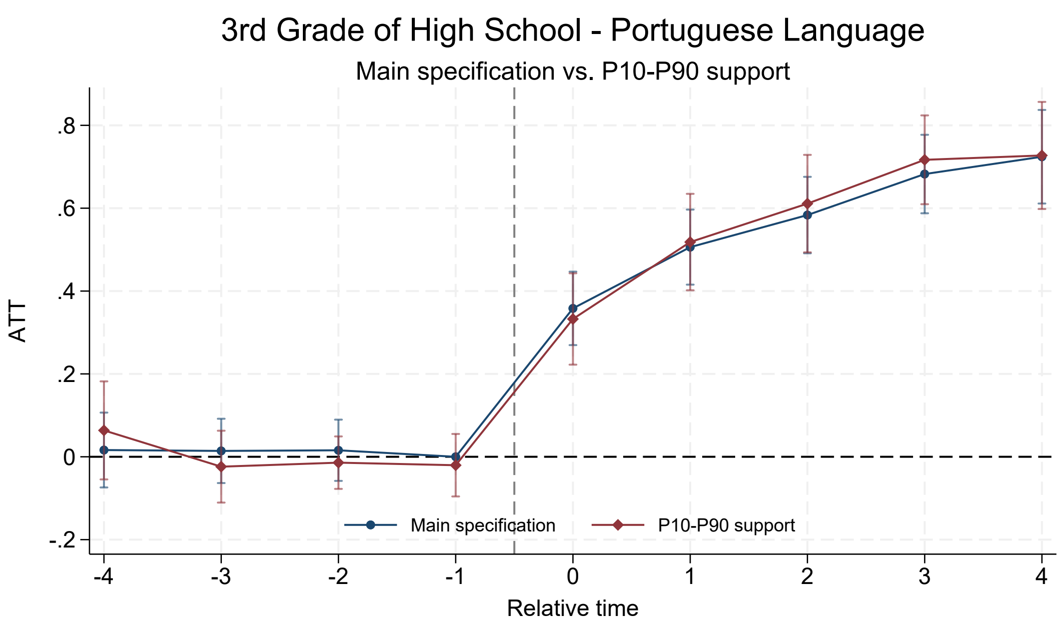
C.2 ATT by Period

Figure 11. Mean-reversion diagnostic: 9th grade, Mathematics



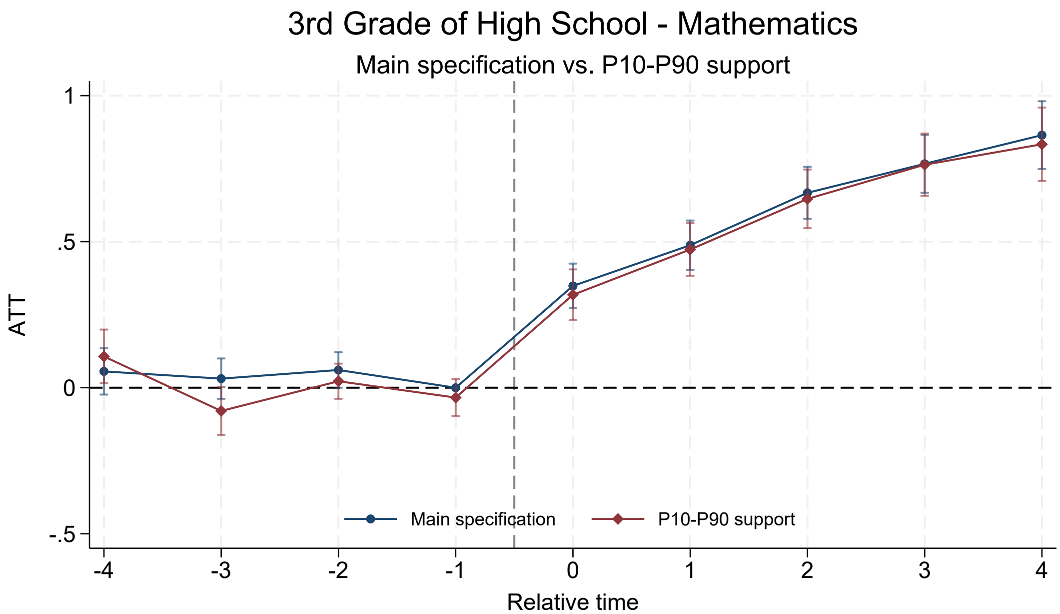
Source: author's elaboration. The restricted line uses schools within the P10–P90 support of baseline proficiency among treated schools.

Figure 12. Mean-reversion diagnostic: 3rd grade of high school, Portuguese Language



Source: author's elaboration. The restricted line uses schools within the P10–P90 support of baseline proficiency among treated schools.

Figure 13. Mean-reversion diagnostic: 3rd grade of high school, Mathematics



Source: author's elaboration. The restricted line uses schools within the P10–P90 support of baseline proficiency among treated schools.

Table 17. Mean-reversion diagnostic, 9th LP: ATT by relative period

Relative period	Main	P10-P90 support
-4	0.019 (0.060)	0.063 (0.057)
-3	0.016 (0.043)	-0.097 (0.043)
-2	0.003 (0.030)	-0.067 (0.042)
-1	0.000	-0.011 (0.034)
0	0.303 (0.039)	0.327 (0.045)
1	0.425 (0.038)	0.426 (0.044)
2	0.528 (0.040)	0.556 (0.044)
3	0.598 (0.046)	0.622 (0.051)
4	0.698 (0.051)	0.734 (0.058)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 18. Mean-reversion diagnostic, 9th Math: ATT by relative period

Relative period	Main	P10-P90 support
-4	0.060 (0.065)	0.023 (0.081)
-3	0.037 (0.040)	-0.040 (0.043)
-2	-0.003 (0.030)	-0.073 (0.041)
-1	0.000	0.006 (0.033)
0	0.310 (0.038)	0.317 (0.043)
1	0.442 (0.036)	0.446 (0.040)
2	0.602 (0.039)	0.627 (0.043)
3	0.751 (0.049)	0.755 (0.055)
4	0.901 (0.062)	0.914 (0.073)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 19. Mean-reversion diagnostic, HS3 LP: ATT by relative period

Relative period	Main	P10-P90 support
-4	0.016 (0.046)	0.064 (0.060)
-3	0.014 (0.040)	-0.024 (0.044)
-2	0.016 (0.038)	-0.014 (0.032)
-1	0.000	-0.020 (0.038)
0	0.358 (0.045)	0.333 (0.056)
1	0.506 (0.046)	0.518 (0.059)
2	0.583 (0.047)	0.611 (0.060)
3	0.682 (0.048)	0.717 (0.055)
4	0.724 (0.058)	0.727 (0.066)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 20. Mean-reversion diagnostic, HS3 Math: ATT by relative period

Relative period	Main	P10-P90 support
-4	0.056 (0.041)	0.107 (0.047)
-3	0.031 (0.035)	-0.080 (0.042)
-2	0.061 (0.031)	0.022 (0.031)
-1	0.000	-0.034 (0.032)
0	0.349 (0.039)	0.318 (0.044)
1	0.488 (0.043)	0.473 (0.046)
2	0.667 (0.045)	0.647 (0.051)
3	0.767 (0.051)	0.763 (0.054)
4	0.865 (0.059)	0.834 (0.064)

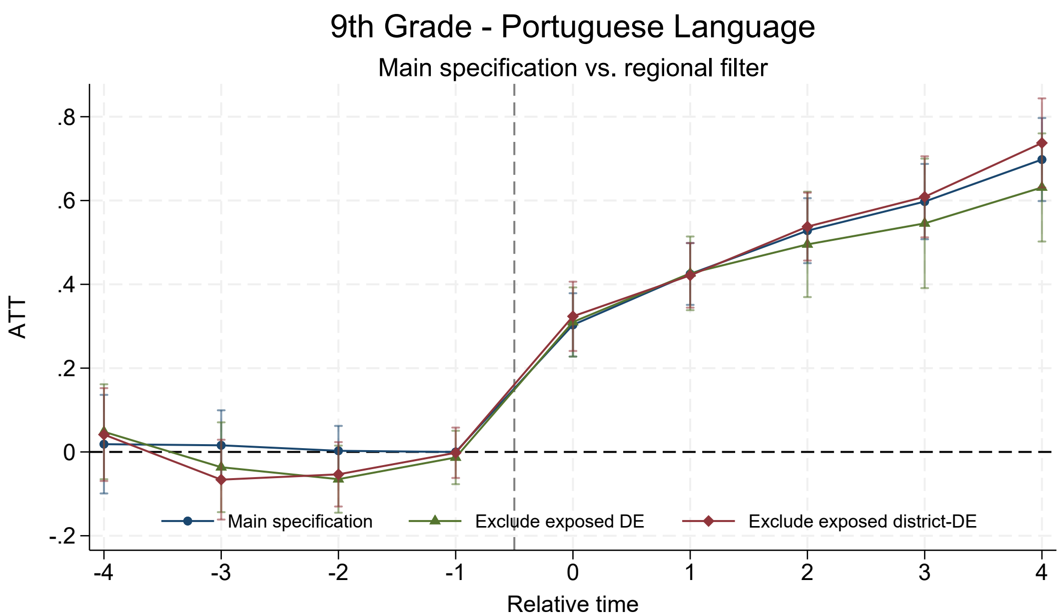
Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

D Robustness to Regional Spillovers

This appendix presents the complete dynamics for specifications that restrict the control group in regions with active PEI schools. The first restriction excludes controls in school districts with at least one active PEI school in the year. The second uses a finer administrative unit formed by the school district and neighborhood.

D.1 Event Studies by Model

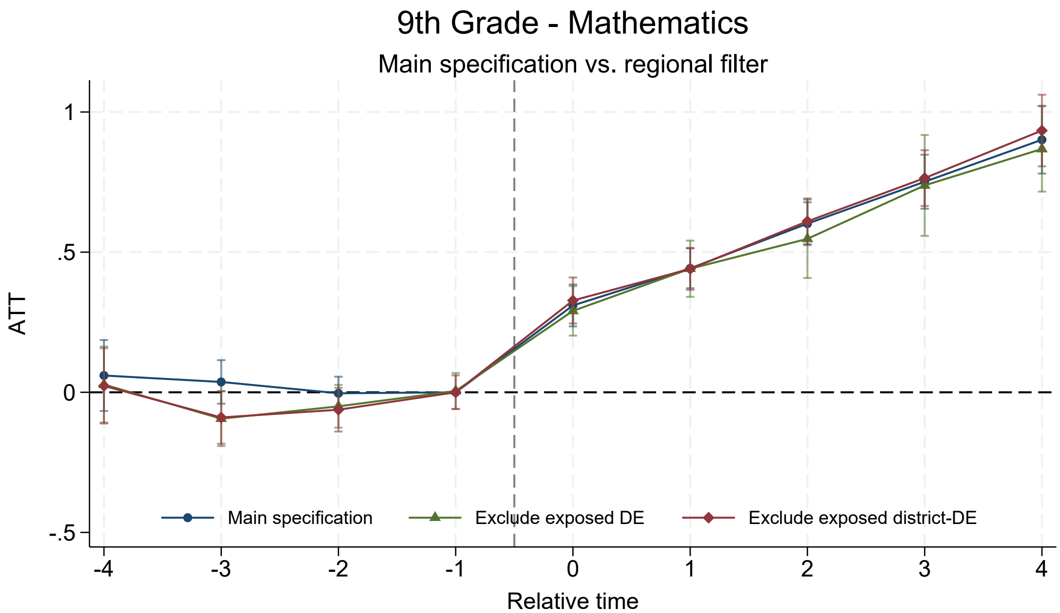
Figure 14. Robustness to regional spillovers: 9th grade, Portuguese Language



Source: author's elaboration. The restrictions exclude controls in administrative units already exposed to active PEI schools.

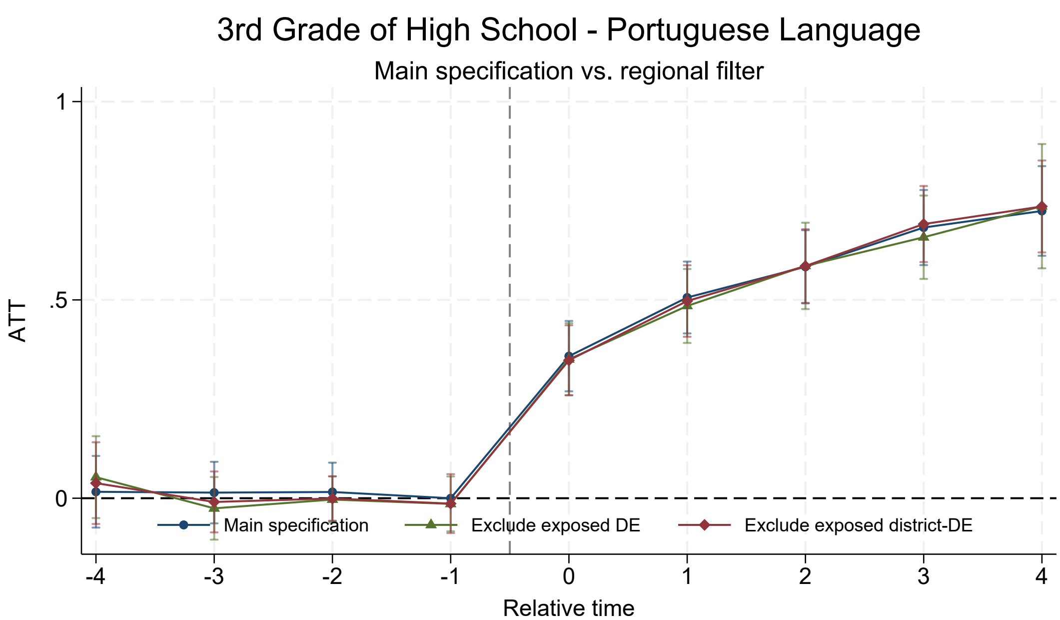
D.2 ATT by Period

Figure 15. Robustness to regional spillovers: 9th grade, Mathematics



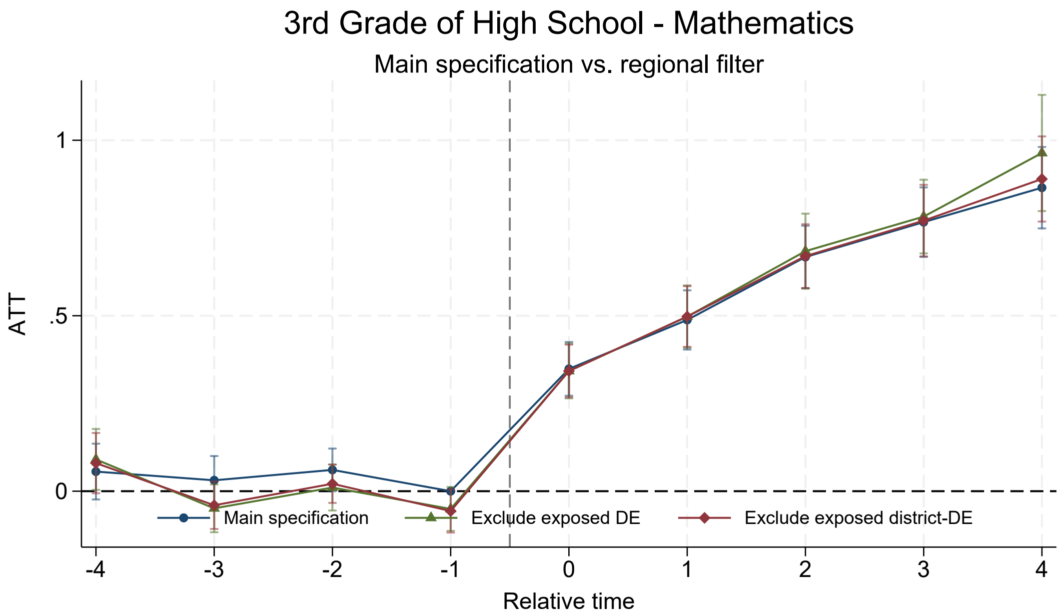
Source: author's elaboration. The restrictions exclude controls in administrative units already exposed to active PEI schools.

Figure 16. Robustness to regional spillovers: 3rd grade of high school, Portuguese Language



Source: author's elaboration. The restrictions exclude controls in administrative units already exposed to active PEI schools.

Figure 17. Robustness to regional spillovers: 3rd grade of high school, Mathematics



Source: author's elaboration. The restrictions exclude controls in administrative units already exposed to active PEI schools.

Table 21. Robustness to regional spillovers, 9th LP: ATT by relative period

Relative period	Main	Exclude exposed DE	Exclude exposed district-DE
-4	0.019 (0.060)	0.048 (0.058)	0.041 (0.057)
-3	0.016 (0.043)	-0.036 (0.055)	-0.066 (0.049)
-2	0.003 (0.030)	-0.065 (0.041)	-0.053 (0.039)
-1	0.000	-0.013 (0.033)	-0.002 (0.031)
0	0.303 (0.039)	0.310 (0.042)	0.324 (0.042)
1	0.425 (0.038)	0.426 (0.045)	0.421 (0.039)
2	0.528 (0.040)	0.495 (0.064)	0.538 (0.041)
3	0.598 (0.046)	0.546 (0.079)	0.609 (0.049)
4	0.698 (0.051)	0.631 (0.066)	0.737 (0.054)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 22. Robustness to regional spillovers, 9th Math: ATT by relative period

Relative period	Main	Exclude exposed DE	Exclude exposed district-DE
-4	0.060 (0.065)	0.028 (0.069)	0.023 (0.069)
-3	0.037 (0.040)	-0.094 (0.050)	-0.090 (0.048)
-2	-0.003 (0.030)	-0.050 (0.039)	-0.062 (0.040)
-1	0.000	0.005 (0.033)	0.000 (0.031)
0	0.310 (0.038)	0.291 (0.045)	0.327 (0.042)
1	0.442 (0.036)	0.441 (0.051)	0.440 (0.038)
2	0.602 (0.039)	0.548 (0.071)	0.610 (0.042)
3	0.751 (0.049)	0.738 (0.092)	0.764 (0.051)
4	0.901 (0.062)	0.868 (0.078)	0.934 (0.065)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 23. Robustness to regional spillovers, HS3 LP: ATT by relative period

Relative period	Main	Exclude exposed DE	Exclude exposed district-DE
-4	0.016 (0.046)	0.053 (0.053)	0.038 (0.053)
-3	0.014 (0.040)	-0.026 (0.040)	-0.009 (0.039)
-2	0.016 (0.038)	-0.003 (0.030)	-0.001 (0.029)
-1	0.000	-0.014 (0.035)	-0.014 (0.038)
0	0.358 (0.045)	0.350 (0.046)	0.348 (0.045)
1	0.506 (0.046)	0.485 (0.048)	0.497 (0.046)
2	0.583 (0.047)	0.586 (0.056)	0.585 (0.047)
3	0.682 (0.048)	0.658 (0.054)	0.691 (0.049)
4	0.724 (0.058)	0.736 (0.080)	0.736 (0.059)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 24. Robustness to regional spillovers, HS3 Math: ATT by relative period

Relative period	Main	Exclude exposed DE	Exclude exposed district-DE
-4	0.056 (0.041)	0.091 (0.044)	0.080 (0.044)
-3	0.031 (0.035)	-0.049 (0.035)	-0.040 (0.035)
-2	0.061 (0.031)	0.010 (0.033)	0.021 (0.028)
-1	0.000	-0.051 (0.032)	-0.057 (0.031)
0	0.349 (0.039)	0.343 (0.040)	0.342 (0.038)
1	0.488 (0.043)	0.497 (0.045)	0.498 (0.044)
2	0.667 (0.045)	0.684 (0.055)	0.670 (0.046)
3	0.767 (0.051)	0.782 (0.054)	0.771 (0.052)
4	0.865 (0.059)	0.964 (0.084)	0.890 (0.062)

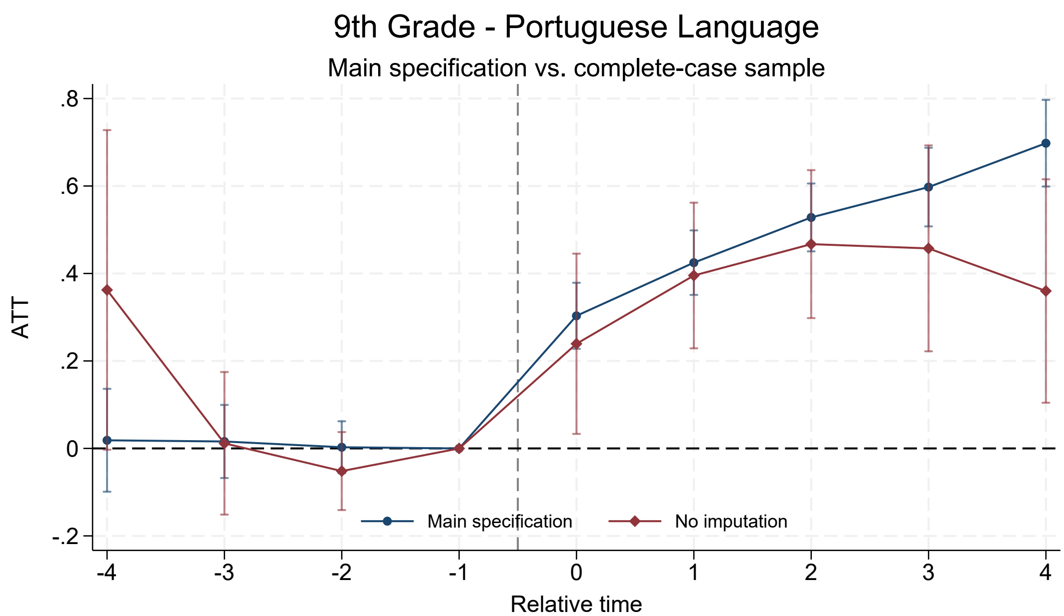
Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

E Imputation and Covariate-Coverage Diagnostic

This appendix details the comparison between the main specification, which imputes composition covariates using annual means when necessary, and the specification with complete raw covariates. In high school, the no-imputation sample becomes too small to produce comparable event studies; for this reason, the tables report the main specification and leave the no-imputation column empty when the estimation is not informative.

E.1 Event Studies by Model

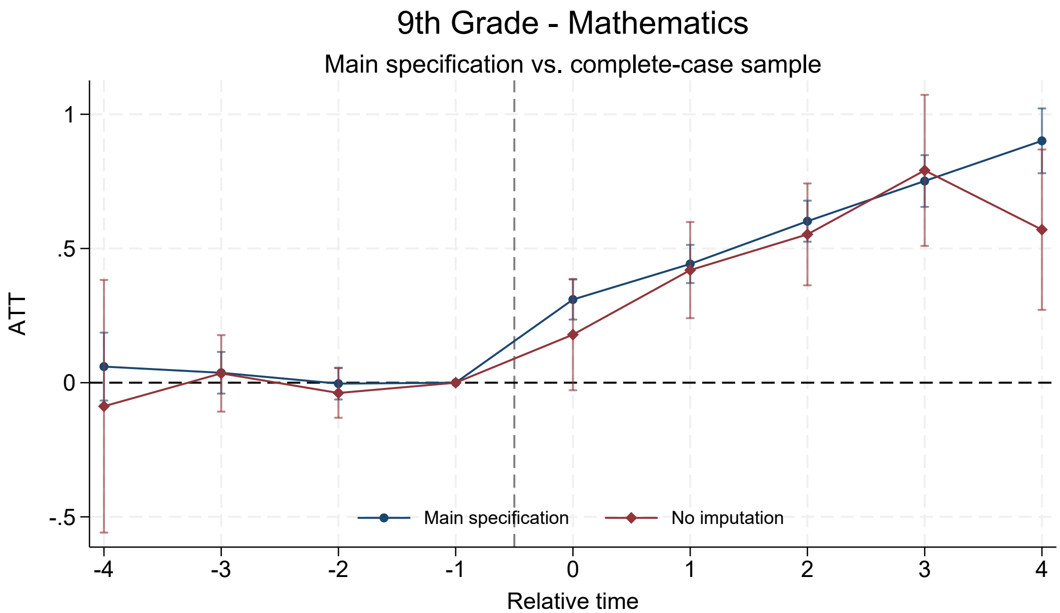
Figure 18. Imputation diagnostic: 9th grade, Portuguese Language



Source: author's elaboration. The no-imputation specification uses only observations with complete raw covariates.

E.2 ATT by Period

Figure 19. Imputation diagnostic: 9th grade, Mathematics



Source: author's elaboration. The no-imputation specification uses only observations with complete raw covariates.

Table 25. Imputation diagnostic, 9th LP: ATT by relative period

Relative period	Main	No imputation
-4	0.019 (0.060)	0.362 (0.186)
-3	0.016 (0.043)	0.012 (0.083)
-2	0.003 (0.030)	-0.052 (0.045)
-1	0.000	0.000
0	0.303 (0.039)	0.239 (0.105)
1	0.425 (0.038)	0.395 (0.085)
2	0.528 (0.040)	0.467 (0.086)
3	0.598 (0.046)	0.457 (0.120)
4	0.698 (0.051)	0.360 (0.130)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 26. Imputation diagnostic, 9th Math: ATT by relative period

Relative period	Main	No imputation
-4	0.060 (0.065)	-0.088 (0.240)
-3	0.037 (0.040)	0.035 (0.073)
-2	-0.003 (0.030)	-0.038 (0.047)
-1	0.000	0.000
0	0.310 (0.038)	0.179 (0.106)
1	0.442 (0.036)	0.419 (0.091)
2	0.602 (0.039)	0.553 (0.097)
3	0.751 (0.049)	0.791 (0.144)
4	0.901 (0.062)	0.570 (0.152)

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 27. Imputation diagnostic, HS3 LP: ATT by relative period

Relative period	Main	No imputation
-4	0.016 (0.046)	–
-3	0.014 (0.040)	–
-2	0.016 (0.038)	–
-1	0.000	–
0	0.358 (0.045)	–
1	0.506 (0.046)	–
2	0.583 (0.047)	–
3	0.682 (0.048)	–
4	0.724 (0.058)	–

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

Table 28. Imputation diagnostic, HS3 Math: ATT by relative period

Relative period	Main	No imputation
-4	0.056 (0.041)	–
-3	0.031 (0.035)	–
-2	0.061 (0.031)	–
-1	0.000	–
0	0.349 (0.039)	–
1	0.488 (0.043)	–
2	0.667 (0.045)	–
3	0.767 (0.051)	–
4	0.865 (0.059)	–

Note: each cell reports the ATT and standard error in parentheses, in standard deviations of annual proficiency. Period $k = -1$ is normalized as the baseline and therefore appears without a standard error.

F Details of the S&X Approximation

This appendix documents the cells used in the aggregate approximation inspired by Sant'Anna and Xu. Table 29 summarizes, by relative period, how many cohort-period cells enter the average and what the average variance is. The following tables show the details by cohort, calendar year, and cell-use condition.

Table 29. S&X approximation: summary of estimated cells by relative period

Grade-subject	k	Cells	Mean ATT	Mean SE	Mean Var.
9th LP	-4	2	0.021	0.072	0.0053
9th LP	-3	3	0.051	0.064	0.0042
9th LP	-2	4	0.020	0.063	0.0041
9th LP	-1	0	–	–	–
9th LP	0	4	0.277	0.094	0.0094
9th LP	1	4	0.408	0.106	0.0119
9th LP	2	4	0.469	0.120	0.0156
9th LP	3	3	0.600	0.131	0.0192
9th LP	4	2	0.755	0.161	0.0307
9th Mat	-4	2	0.061	0.081	0.0074
9th Mat	-3	3	0.054	0.068	0.0049
9th Mat	-2	4	0.010	0.063	0.0041
9th Mat	-1	0	–	–	–
9th Mat	0	4	0.317	0.102	0.0116
9th Mat	1	4	0.444	0.111	0.0130
9th Mat	2	4	0.593	0.138	0.0205
9th Mat	3	3	0.774	0.154	0.0253
9th Mat	4	2	0.898	0.179	0.0352
HS3 LP	-4	1	0.044	0.089	0.0078
HS3 LP	-3	2	0.093	0.088	0.0078
HS3 LP	-2	3	0.035	0.070	0.0050
HS3 LP	-1	0	–	–	–
HS3 LP	0	4	0.434	0.151	0.0241
HS3 LP	1	4	0.547	0.157	0.0280
HS3 LP	2	4	0.629	0.172	0.0340
HS3 LP	3	3	0.648	0.202	0.0500
HS3 LP	4	3	0.677	0.188	0.0398
HS3 Mat	-4	1	0.075	0.077	0.0059
HS3 Mat	-3	2	0.074	0.077	0.0064
HS3 Mat	-2	3	0.073	0.063	0.0041
HS3 Mat	-1	0	–	–	–
HS3 Mat	0	4	0.389	0.140	0.0212
HS3 Mat	1	4	0.551	0.154	0.0281
HS3 Mat	2	4	0.720	0.184	0.0385
HS3 Mat	3	3	0.733	0.204	0.0483
HS3 Mat	4	3	0.833	0.208	0.0484

Note: the number of cells corresponds to cohort-period combinations with enough support to enter the approximate S&X average. ATT, standard error, and variance are simple averages across the cells estimated in each relative period.

F.1 Cells by Cohort and Period

Table 30. S&X approximation, 9th LP: cohort-period cells used in aggregation

k	Cohort	Year	N	Treated N	ATT	SE	Var.	Use
-4	2015	2011	6968	45	0.134	0.062	0.0039	Estimated
-4	2016	2012	6640	28	-0.093	0.082	0.0067	Estimated
-4	2017	2013	6932	6	–	–	–	No support
-3	2014	2011	7053	86	0.121	0.053	0.0028	Estimated
-3	2015	2012	6993	46	0.082	0.067	0.0045	Estimated
-3	2016	2013	6658	28	-0.051	0.073	0.0053	Estimated
-3	2017	2014	6964	6	–	–	–	No support
-2	2013	2011	6958	37	0.106	0.073	0.0053	Estimated
-2	2014	2012	7082	87	0.095	0.051	0.0026	Estimated
-2	2015	2013	7017	48	0.000	0.058	0.0033	Estimated
-2	2016	2014	6694	28	-0.120	0.071	0.0050	Estimated
-2	2017	2015	6842	5	–	–	–	No support
-1	2012	2011	.	.	–	–	–	Base
-1	2013	2012	.	.	–	–	–	Base
-1	2014	2013	.	.	–	–	–	Base
-1	2015	2014	.	.	–	–	–	Base
-1	2016	2015	.	.	–	–	–	Base
-1	2017	2016	.	.	–	–	–	Base
0	2012	2012	6869	0	–	–	–	Small cell
0	2013	2013	6955	21	0.304	0.137	0.0186	Estimated
0	2014	2014	7074	77	0.302	0.068	0.0047	Estimated
0	2015	2015	6872	43	0.331	0.086	0.0074	Estimated
0	2016	2016	6700	22	0.172	0.084	0.0071	Estimated
0	2017	2017	6992	0	–	–	–	Small cell
1	2012	2013	6854	0	–	–	–	Small cell
1	2013	2014	6935	21	0.429	0.147	0.0215	Estimated
1	2014	2015	6898	77	0.426	0.076	0.0057	Estimated
1	2015	2016	7004	42	0.426	0.099	0.0098	Estimated
1	2016	2017	6669	22	0.352	0.104	0.0108	Estimated
1	2017	2018	6973	0	–	–	–	Small cell
2	2012	2014	6838	0	–	–	–	Small cell
2	2013	2015	6762	20	0.484	0.175	0.0305	Estimated
2	2014	2016	7033	75	0.548	0.081	0.0065	Estimated
2	2015	2017	6975	42	0.577	0.119	0.0141	Estimated
2	2016	2018	6659	22	0.268	0.107	0.0114	Estimated
3	2012	2015	6667	0	–	–	–	Small cell
3	2013	2016	6901	21	0.665	0.196	0.0383	Estimated
3	2014	2017	7005	75	0.632	0.093	0.0087	Estimated
3	2015	2018	6957	43	0.504	0.103	0.0107	Estimated
4	2012	2016	6805	0	–	–	–	Small cell
4	2013	2017	6869	21	0.862	0.230	0.0531	Estimated
4	2014	2018	6984	76	0.648	0.092	0.0084	Estimated

Note: *Estimated* indicates a cell used in the S&X average; *No support* indicates low overlap in the multinomial logit; *Small cell* indicates no treated units or an insufficient number of treated units; *Base* is the normalized period $k = -1$.

Table 31. S&X approximation, 9th Math: cohort-period cells used in aggregation

k	Cohort	Year	N	Treated N	ATT	SE	Var.	Use
-4	2015	2011	6968	45	0.125	0.051	0.0026	Estimated
-4	2016	2012	6670	28	-0.004	0.110	0.0122	Estimated
-4	2017	2013	6932	6	–	–	–	No support
-3	2014	2011	7053	86	0.112	0.056	0.0031	Estimated
-3	2015	2012	6993	46	0.095	0.058	0.0033	Estimated
-3	2016	2013	6688	28	-0.046	0.091	0.0083	Estimated
-3	2017	2014	6964	6	–	–	–	No support
-2	2013	2011	6958	37	0.057	0.073	0.0053	Estimated
-2	2014	2012	7082	87	0.057	0.056	0.0031	Estimated
-2	2015	2013	7017	48	0.008	0.055	0.0030	Estimated
-2	2016	2014	6724	28	-0.083	0.069	0.0048	Estimated
-2	2017	2015	6857	5	–	–	–	No support
-1	2012	2011	.	.	–	–	–	Base
-1	2013	2012	.	.	–	–	–	Base
-1	2014	2013	.	.	–	–	–	Base
-1	2015	2014	.	.	–	–	–	Base
-1	2016	2015	.	.	–	–	–	Base
-1	2017	2016	.	.	–	–	–	Base
0	2012	2012	6869	0	–	–	–	Small cell
0	2013	2013	6955	21	0.385	0.160	0.0257	Estimated
0	2014	2014	7074	77	0.294	0.068	0.0047	Estimated
0	2015	2015	6887	43	0.334	0.078	0.0061	Estimated
0	2016	2016	6730	22	0.257	0.100	0.0099	Estimated

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k	Cohort	Year	N	Treated N	ATT	SE	Var.	Use
0	2017	2017	6992	0	–	–	–	Small cell
1	2012	2013	6854	0	–	–	–	Small cell
1	2013	2014	6935	21	0.465	0.146	0.0212	Estimated
1	2014	2015	6913	77	0.449	0.075	0.0057	Estimated
1	2015	2016	7004	42	0.417	0.091	0.0082	Estimated
1	2016	2017	6699	22	0.446	0.131	0.0171	Estimated
1	2017	2018	6973	0	–	–	–	Small cell
2	2012	2014	6838	0	–	–	–	Small cell
2	2013	2015	6777	20	0.564	0.188	0.0354	Estimated
2	2014	2016	7033	75	0.599	0.088	0.0078	Estimated
2	2015	2017	6975	42	0.590	0.110	0.0120	Estimated
2	2016	2018	6689	22	0.620	0.164	0.0270	Estimated
3	2012	2015	6682	0	–	–	–	Small cell
3	2013	2016	6901	21	0.736	0.201	0.0406	Estimated
3	2014	2017	7005	75	0.696	0.102	0.0104	Estimated
3	2015	2018	6957	43	0.889	0.158	0.0250	Estimated
4	2012	2016	6805	0	–	–	–	Small cell
4	2013	2017	6869	21	0.882	0.235	0.0551	Estimated
4	2014	2018	6984	76	0.914	0.123	0.0152	Estimated

Note: *Estimated* indicates a cell used in the S&X average; *No support* indicates low overlap in the multinomial logit; *Small cell* indicates no treated units or an insufficient number of treated units; *Base* is the normalized period $k = -1$.

Table 32. S&X approximation, HS3 LP: cohort-period cells used in aggregation

k	Cohort	Year	N	Treated N	ATT	SE	Var.	Use
-4	2015	2011	6727	39	–	–	–	No support
-4	2016	2012	6455	23	0.044	0.089	0.0078	Estimated
-4	2017	2013	6791	8	–	–	–	No support
-3	2014	2011	6797	76	0.128	0.077	0.0060	Estimated
-3	2015	2012	6772	39	–	–	–	No support
-3	2016	2013	6495	24	0.059	0.098	0.0096	Estimated
-3	2017	2014	6825	8	–	–	–	No support
-2	2013	2011	6711	42	0.039	0.065	0.0042	Estimated
-2	2014	2012	6846	79	0.056	0.071	0.0051	Estimated
-2	2015	2013	6808	39	–	–	–	No support
-2	2016	2014	6532	25	0.009	0.075	0.0056	Estimated
-2	2017	2015	6701	7	–	–	–	No support
-1	2012	2011	.	.	–	–	–	Base
-1	2013	2012	.	.	–	–	–	Base
-1	2014	2013	.	.	–	–	–	Base
-1	2015	2014	.	.	–	–	–	Base
-1	2016	2015	.	.	–	–	–	Base
-1	2017	2016	.	.	–	–	–	Base
0	2012	2012	6611	10	0.340	0.209	0.0435	Estimated
0	2013	2013	6730	28	0.570	0.153	0.0234	Estimated
0	2014	2014	6844	54	0.375	0.107	0.0114	Estimated
0	2015	2015	6677	29	–	–	–	No support
0	2016	2016	6549	21	0.451	0.135	0.0181	Estimated
0	2017	2017	6893	8	–	–	–	No support
1	2012	2013	6598	10	0.580	0.247	0.0609	Estimated
1	2013	2014	6719	30	0.695	0.169	0.0287	Estimated
1	2014	2015	6693	60	0.506	0.099	0.0098	Estimated
1	2015	2016	6822	30	–	–	–	No support
1	2016	2017	6548	22	0.408	0.112	0.0126	Estimated
1	2017	2018	6861	8	–	–	–	No support
2	2012	2014	6586	10	0.631	0.282	0.0795	Estimated
2	2013	2015	6557	28	0.703	0.163	0.0265	Estimated
2	2014	2016	6840	61	0.667	0.113	0.0128	Estimated
2	2015	2017	6819	29	–	–	–	No support
2	2016	2018	6521	22	0.517	0.132	0.0173	Estimated
3	2012	2015	6426	9	0.553	0.333	0.1108	Estimated
3	2013	2016	6707	29	0.750	0.168	0.0282	Estimated
3	2014	2017	6841	62	0.641	0.105	0.0111	Estimated
3	2015	2018	6788	28	–	–	–	No support
4	2012	2016	6577	10	0.524	0.280	0.0784	Estimated
4	2013	2017	6710	29	0.778	0.166	0.0276	Estimated
4	2014	2018	6812	61	0.728	0.117	0.0136	Estimated

Note: *Estimated* indicates a cell used in the S&X average; *No support* indicates low overlap in the multinomial logit; *Small cell* indicates no treated units or an insufficient number of treated units; *Base* is the normalized period $k = -1$.

Table 33. S&X approximation, HS3 Math: cohort-period cells used in aggregation

k	Cohort	Year	N	Treated N	ATT	SE	Var.	Use
-4	2015	2011	6727	39	–	–	–	No support
-4	2016	2012	6485	23	0.075	0.077	0.0059	Estimated
-4	2017	2013	6791	8	–	–	–	No support
-3	2014	2011	6797	76	0.107	0.057	0.0033	Estimated
-3	2015	2012	6772	39	–	–	–	No support
-3	2016	2013	6525	24	0.041	0.097	0.0095	Estimated
-3	2017	2014	6825	8	–	–	–	No support
-2	2013	2011	6711	42	0.077	0.078	0.0060	Estimated
-2	2014	2012	6846	79	0.108	0.060	0.0037	Estimated
-2	2015	2013	6808	39	–	–	–	No support
-2	2016	2014	6562	25	0.035	0.050	0.0025	Estimated
-2	2017	2015	6716	7	–	–	–	No support
-1	2012	2011	.	.	–	–	–	Base
-1	2013	2012	.	.	–	–	–	Base
-1	2014	2013	.	.	–	–	–	Base
-1	2015	2014	.	.	–	–	–	Base
-1	2016	2015	.	.	–	–	–	Base
-1	2017	2016	.	.	–	–	–	Base
0	2012	2012	6611	10	0.227	0.197	0.0387	Estimated
0	2013	2013	6730	28	0.585	0.150	0.0226	Estimated
0	2014	2014	6844	54	0.363	0.094	0.0088	Estimated
0	2015	2015	6692	29	–	–	–	No support
0	2016	2016	6579	21	0.383	0.121	0.0146	Estimated
0	2017	2017	6893	8	–	–	–	No support
1	2012	2013	6598	10	0.571	0.259	0.0673	Estimated
1	2013	2014	6719	30	0.699	0.160	0.0257	Estimated
1	2014	2015	6708	60	0.531	0.094	0.0089	Estimated
1	2015	2016	6822	30	–	–	–	No support
1	2016	2017	6578	22	0.402	0.103	0.0107	Estimated
1	2017	2018	6861	8	–	–	–	No support
2	2012	2014	6586	10	0.703	0.300	0.0898	Estimated
2	2013	2015	6572	28	0.798	0.171	0.0292	Estimated
2	2014	2016	6840	61	0.796	0.122	0.0149	Estimated
2	2015	2017	6819	29	–	–	–	No support
2	2016	2018	6551	22	0.583	0.142	0.0202	Estimated
3	2012	2015	6442	10	0.563	0.314	0.0984	Estimated
3	2013	2016	6707	29	0.869	0.182	0.0330	Estimated
3	2014	2017	6841	62	0.766	0.116	0.0135	Estimated
3	2015	2018	6788	28	–	–	–	No support
4	2012	2016	6577	10	0.699	0.307	0.0944	Estimated
4	2013	2017	6710	29	0.880	0.180	0.0323	Estimated
4	2014	2018	6812	61	0.921	0.136	0.0184	Estimated

Note: *Estimated* indicates a cell used in the S&X average; *No support* indicates low overlap in the multinomial logit; *Small cell* indicates no treated units or an insufficient number of treated units; *Base* is the normalized period $k = -1$.